

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING

Kathmandu Engineering College
Department of Computer Engineering



Major Project Final Report On

TrashNet

[Code Number: CT755]

By:

Pawan Tiwari	KAT078BCT056
Pratik Gurung	KAT078BCT059
Sudyumna Mishra	KAT078BCT083
Sujan Malakar	KAT078BCT085

TO

DEPARTMENT OF COMPUTER ENGINEERING

KATHMANDU, NEPAL

Chaitra, 2082

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
KATHMANDU ENGINEERING COLLEGE
DEPARTMENT OF COMPUTER ENGINEERING



Major Project Final Report On

TrashNet

[CODE NUMBER: CT 755]

PROJECT REPORT SUBMITTED TO THE DEPARTMENT OF
COMPUTER ENGINEERING IN PARTIAL FULFILLMENT OF
REQUIREMENTS FOR THE
BACHELOR OF ENGINEERING

BY

Pawan Tiwari	KAT078BCT056
Pratik Gurung	KAT078BCT059
Sudyumna Mishra	KAT078BCT083
Sujan Malakar	KAT078BCT085

KATHMANDU, NEPAL

Chaitra, 2082

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to the Department of Computer Engineering, Kathmandu Engineering College for providing us such a great opportunity to undertake this project. This project has helped us to develop our skills which will greatly benefit us in our future.

We would like to express our special thanks of gratitude to **Assoc. Prof. Sudeep Shakya** (Head of Department), **Er. Kunjan Amatya** (Deputy head of Department) of Computer Engineering Department.

We deeply express our sincere thanks to **Er. Sharad Chandra Joshi** (Project Coordinator) and **Er. Ritu Bajracharya** (Project Supervisor) for their valuable suggestions, encouragement and recommendation for developing this project. Finally, we would like to extend our thanks to all the teachers and team members who helped with their valuable suggestions and cooperation in every aspect of this project. Any comments and suggestions about the improvement are always appreciated.

ABSTRACT

Conventional dustbins face major challenges in today's urban and domestic environments leading to improper waste segregation, resulting in environmental pollution, difficulty in recycling, and increased landfill burden. Users also don't have real-time information about the fill levels, causing bins to overflow, emit foul odors, and attract pests. This project outlines an AI-based Smart Bin System capable of automatically segregating waste into organic, plastic, paper, metal, and unknown categories. Each dustbin is equipped with ESP32-CAM to capture image of the waste and send it to the AI model to categorize waste type, servo motors to dispose waste to proper compartments, and ultrasonic sensors to monitor the compartment fill level. The AI model is trained using combination of Garbage Classification (12 classes) by Mostafa Mohamed, Garbage Dataset by Suman Kunwar and some custom datasets. Home users can monitor their dustbin through a mobile application: TrashGold, while public dustbins are monitored via a desktop website: TrashNet, which also keeps information of bin analytics and its map location, generates QR codes, rewards users with points, and displays a leaderboard. This integrated system promotes efficient waste management, reduces manual labor, improve waste segregation accuracy, and optimize disposal processes through smart automation. The model achieves an overall accuracy of approximately 91%, demonstrating strong and reliable performance in classifying waste into different categories.

Keywords: *Smart Bin, waste type identification, waste segregation, fill-level monitoring, IoT, real-time notification, sustainable waste management.*

TABLE OF CONTENTS

ACKNOWLEDGEMENT	iii
ABSTRACT.....	iv
LIST OF FIGURES	vii
LIST OF TABLES.....	viii
LIST OF ABBREVIATIONS.....	ix
CHAPTER 1: INTRODUCTION	1
1.1 BACKGROUND THEORY	1
1.2 PROBLEM STATEMENT.....	5
1.3 OBJECTIVE	5
1.4 SCOPE AND APPLICATIONS.....	6
CHAPTER 2: LITERATURE REVIEW	7
2.1 EXISTING SYSTEMS	12
2.2 LIMITATIONS OF EXISTING SYSTEM.....	13
2.3 SOLUTIONS PROVIDED	13
CHAPTER 3: METHODOLOGY	14
3.1 PROCESS MODEL	14
3.2. SYSTEM BLOCK DIAGRAM	17
3.3 FLOWCHARTS.....	20
3.3.1 FLOWCHART OF WASTE DISPOSAL PROCESS	20
3.3.2 FLOWCHART OF MOBILE APPLICATION TRASHGOLD	21
3.3.3 FLOWCHART OF DESKTOP WEBSITE TRASHNET	22
3.4 ALGORITHM.....	23
3.4.1 CONVOLUTIONAL NEURAL NETWORK (CNN)	23
3.5 UML DIAGRAMS.....	30
3.5.1 USE CASE DIAGRAM	30

3.5.2 SEQUENCE DIAGRAM	31
3.5.3 CLASS DIAGRAM.....	32
3.6 VERIFICATION AND VALIDATION	33
3.6.1 VERIFICATION	33
3.6.2 VALIDATION	34
3.7 TOOLS USED.....	38
3.7.1 SOFTWARE TOOLS.....	38
3.7.2 HARDWARE TOOLS	39
3.8 DATASET COLLECTION	41
CHAPTER 4: EPILOGUE.....	42
4.1 RESULT.....	42
4.2 CONCLUSION	43
4.3 FUTURE ENHANCEMENT.....	44
REFERENCES	46
SCREENSHOTS.....	48

LIST OF FIGURES

Figure 1:Classification of Organic Waste.....	2
Figure 2:Classification of Inorganic Waste	3
Figure 3: Iterative Waterfall Model	14
Figure 4: System Block Diagram.....	17
Figure 5: Flowchart of Waste disposal process in the bin	20
Figure 6: Flowchart of mobile application TrashGold	21
Figure 7: Flowchart of desktop website TrashNet.....	22
Figure 8: CNN model.....	24
Figure 9: ReLU Function Graph.....	26
Figure 10: Simplified CNN (Convolutional Neural Network) architecture	28
Figure 11: Use Case Diagram of TrashNet.....	30
Figure 12: Sequence Diagram of TrashNet	31
Figure 13: Class Diagram of TrashNet.....	32
Figure 14: Sample of Dataset use for AI Model Training	33
Figure 15: Training and Validation Loss	34
Figure 16: Model Detection Performance During Validation	35
Figure 17: Confusion Matrix of the Model.....	42
Figure 18: F1-Confidence Curve of the Model.....	43

LIST OF TABLES

Table 1: Class labels and names	34
Table 2: Anamolies and their potential causes	36

LIST OF ABBREVIATIONS

API	:	Application Programming Interface
AWS	:	Amazon Web Services
GSM	:	Global System for Mobile Communication
IDE	:	Integrated Development Environment
IoT	:	Internet of Things
ISSWMS	:	Intelligent and Smart Solid Waste Management System
LoRa	:	Long Range
MWBMS	:	Manual Waste Bin Management System
OpenCV	:	Open Source Computer Vision Library
PIR	:	Passive Infrared
QR	:	Quick Response
RGB	:	Red Green Blue
SWBMS	:	Smart Waste Bin Management System
UI	:	User Interface
Wifi	:	Wireless Fidelity

CHAPTER 1: INTRODUCTION

1.1 BACKGROUND THEORY

The world is facing a rapidly growing waste management crisis mainly due to rapid urbanization and population growth, leading to an unprecedented increase in solid waste generation. According to global reports, the world produces over 2 billion tons of municipal solid waste annually, and this figure is expected to rise sharply in the coming years. Improper disposal, lack of recycling, and reliance on landfills contribute to environmental pollution, greenhouse gas emissions, and public health hazards. Nepal is no exception; with increasing urban populations, waste generation has risen significantly, while existing waste management infrastructure struggles to handle this increasing volume of waste. Unregulated dumping, open burning, and lack of effective recycling methods have resulted in overflowing landfills, blocked drainage systems, and polluted rivers, posing serious risks to public health and the environment.

Traditionally, conventional waste bins or manual waste bins have been a primary method of waste collection. They are simple containers designed to temporarily store solid waste until it is collected and transported to disposal sites, such as landfills or dumping grounds. However, with rapid urbanization, population growth, and changing consumption patterns, the volume and variety of waste have increased significantly. Conventional bins do not support waste segregation, fill-level detection, or monitoring, which are essential concepts in modern solid waste management that emphasizes reduction, reuse, recycling, and sustainable disposal.

Due to the lack of intelligence and communication capability, conventional bins suffer from several practical problems. Waste often gets mixed, making recycling difficult and increasing the burden on landfills. Since there is no real-time information about bin status, they may overflow or remain uncollected for long periods, leading to unhygienic conditions, foul odors, and the spread of diseases. Collection processes are time-based rather than need-based, resulting in inefficient use of manpower and fuel. Additionally, the absence of data generation prevents authorities from analyzing waste patterns and improving planning. These limitations reduce overall efficiency and environmental

sustainability, which smart bins aim to overcome by integrating sensors, automation, and intelligent monitoring.

For simplification, we classify the wastes into 2 categories:

Organic Waste: Organic waste is any material that is biodegradable and comes from either a plant or an animal. These waste is organic material that can be broken into carbon dioxide, methane or simple organic molecules. Examples of organic waste include green waste, food waste, food-soiled paper, non-hazardous wood waste, green waste, and landscape and pruning waste.

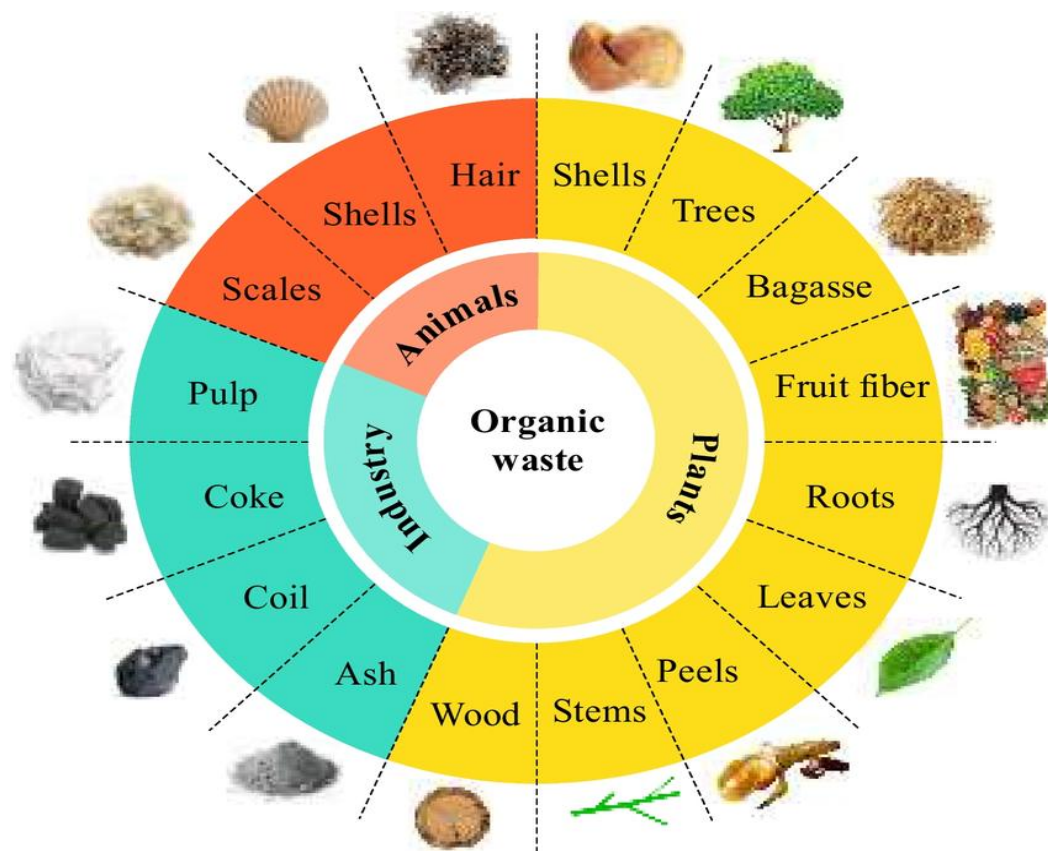


Figure 1: Classification of Organic Waste [1]

Inorganic waste: Inorganic waste is a type of waste that does not contain organic compounds. This waste is generally very difficult to decompose by microorganisms. Glass, aluminum cans, dust, and metal are some examples of inorganic waste. Inorganic waste remains free from decay, with more than 500 years needed being common for effective decomposition, therefore disposal can be challenging. Reducing consumption, reusing, and recycling are possible solutions for coping with this type of waste. Papers,

Plastic and Metal in particular take long time to decay and need to be recycled for clean environment.



Figure 2:Classification of Inorganic Waste [2]

To address these challenges, smart waste management systems have emerged, leveraging Internet of Things (IoT), machine learning, and automation to optimize waste collection, segregation, and disposal. A Smart Bin with automatic waste type identification, segregation, and real-time fill-level monitoring can significantly improve recycling efficiency, reduce operational costs, and minimize environmental impact.

Waste Type Identification Technologies

Sensor-Based Waste Detection

The classification of waste into categories (e.g., organic, inorganic, recyclable) can be achieved using:

- A high-resolution RGB camera captures images of deposited waste.
- Preprocessing techniques (resizing, normalization, noise reduction) enhance image quality before classification.
- Real-time image processing ensures quick identification (critical for user-friendly operation).

Image Processing & Machine Learning: Cameras combined with AI (e.g., Convolutional Neural Networks) can classify waste based on visual features.

Machine Learning for Waste Classification

Supervised Learning Models (CNN) can be trained on labeled waste datasets.

Deep Learning Approaches (e.g., TensorFlow, YOLO for real-time object detection) improve accuracy in dynamic environments.

Automatic Waste Segregation Mechanism

Once waste is classified, an automated segregation system directs it to the appropriate compartment using:

- Servo Motors / Actuators: To open and close designated bin compartments.
- Conveyor Belt Systems: For moving waste to the correct section (in larger bins).
- Rotary Mechanisms: For smaller bins with multiple internal partitions.

Fill-Level Monitoring Using Ultrasonic Sensors

Ultrasonic Sensors measure the distance between the sensor and the waste surface to estimate fill level.

IoT Integration: Data is transmitted to a cloud server (e.g., Firebase, AWS IoT) and displayed on a mobile app.

Alert System: Notifications are sent when the bin reaches a predefined threshold (e.g., 80% full).

IoT and Cloud Connectivity

Microcontroller (Arduino) processes sensor data.

Wireless Communication (Wi-Fi/Bluetooth/LoRa) sends data to a cloud platform.

Mobile App Interface provides real-time bin status, historical data, and collection alerts.

1.2 PROBLEM STATEMENT

Traditional dustbins don't have the capability to identify what type of wastage is disposed in it, causing it to heavily rely on people's knowledge of the nature of wastage. This often leads to incorrect disposal, either due to negligence or lack of knowledge. Additionally, there is no way to notice if the Traditional dustbin has reached its maximum storage capacity without having to manually check it. This could eventually lead to waste overflow which causes bad odor and attracts pests, which ultimately harms the hygiene standards in the area.

1.3 OBJECTIVE

- To develop a Smart Bin system that automatically recognizes, categorizes, and segregates waste while providing timely updates on its storage status.

1.4 SCOPE AND APPLICATIONS

1. Personal Hygiene

The Smart Bin uses storage capacity sensors to notify users when a compartment is full, reducing waste overflow and improving environmental and personal hygiene.

2. Hospitals

The Smart Bin can assist in automatically segregating hazardous waste and non-hazardous waste, improving safety and following medical waste regulations.

3. Municipal waste handling

Data collected from multiple Smart Bins can help city planners identify waste generation patterns across different areas for more effective municipal waste handling.

4. Waste Management Companies

Storage and waste-type data from Smart Bins can help optimize collection routes, reducing fuel usage and operational costs.

5. Parks and Public Spaces

By simplifying the disposal process and encouraging proper waste segregation, Smart Bins can increase public participation in cleanliness efforts and reduce littering.

6. Recycling Centers

Recycling Centers can use Smart Bins to pre-sort wastage, reducing the manual labor and improving their efficiency.

CHAPTER 2: LITERATURE REVIEW

With an increasing amount of waste and inefficiency of conventional waste management systems which are prone to errors, there is a need of smart waste management systems. The smart bins with AI and IoT have emerged as an innovative solution which identify and segregate waste in real time and provide real-time data.

The project titled “Waste Segregation System” Roshan Khadka, Priyanshu Joshi, Nandan Kumar Kanth, Sujin Gwachha focused on developing an effective waste segregation system for automatic waste classification. For this, they developed a model for waste classification using VGG-16 which classified the waste into paper, metal and plastic types. In their system, the controller detected the presence of waste using an IR sensor. The picture of waste to be categorized was preprocessed and sent to the VGG-16 model, which was trained using about 1750 images of paper and plastic and 1500 images of metal. The model selected the category with the highest prediction as the predicted category of the input image. Then, signal was sent to Raspberry Pi which then sent the signal to the servo motor to rotate the bin accordingly. They used two bins: one was divided into three compartments for categorizing three different categories while the other was used as excess bin where user threw the waste when first bin was full which was displayed by the LCD. The model achieved an accuracy of 85.76% for validation test and 85.44% for the test dataset. The precision, recall, and F1-score values of the model also indicated a strong performance across all three waste categories, with paper exhibiting the highest recall of 0.93 in validation and 0.94 in testing [3].

The paper titled “Waste Segregation Using Artificial Intelligence” by Sindhu Rajendran, Vidhya Shree, Rajat Keshri, Rohit S, Rachana S presented a non-manual waste segregation process as the current manual waste segregation method is not effective in terms of accuracy and time interval required. They developed a simple and effective waste segregation system using machine learning that segregated the waste into cardboard, glass, metal, trash, paper and plastic. For this, they collected about 500 images of cardboard, glass, paper, plastic, metal and random trash (soda cans). They changed each of these images into same dimension and converted the color into grayscale and created output labels for each dataset. The images are then converted and stored in a 1D array which are further used for labeling while testing. By using CNN,

the input images were sampled and then convolved to determine the edges in the images. The CNN model was then trained on the training set and validated on the test set. With the given small dataset, and the basic convolution model, they got an accuracy of about 75% on test case and 65% on validation case demonstrating the success in completely removing human intervention in the waste segregation stage [4].

A study titled “Smart Bin and IoT: A Sustainable Future for Waste Management System in Nigeria” by Ayodeji Okubanjo, Bashir Odufuwa, Alexander Okandeji and Emmanuel Daniel presented a Smart Waste Bin Management System (SWBMS) to address the issues regarding environmental and health crises caused by Nigeria's waste collection and disposal problems. They developed a Smart Waste Bin Management System (SWBMS) with efficient Internet of Things (IoT) which provided a sustainable solution to address the above issues in the Nigerian cities. They used ultrasonic sensors, HC-SR04 for identification of person and waste level in bin, a WI-Fi ESP8266 module for internet connectivity, an Arduino Uno, 16 X2 LCD, a servo motor for lid system control, a buzzer for alarm notification, and a waste bin. They compared their system with traditional manual waste bin management system (MWBMS) currently used in typical Nigerian cities and found the system evaluation statement as "Excellent" from 85% of users and "Good" from 15% which demonstrated that implementing this system would serve as a roadmap for the development of waste management systems that can improve city hygiene. They also compared the costs required to develop the prototype of the system with the cost spent by administration of their university for cleaning staffs and found a huge difference in those costs demonstrating the cost-effectiveness and improved time and energy efficiency of the proposed system [5].

Anuradha Singh and Prabhakar Tiwari in their project titled “Smart Dustbin using IoT notifications” focused on the challenges created due to overflowing of dustbins located in public places in India which generated an unsanitary environment and a foul odor. They presented a simple, cost-effective and time-saving solution by employing a mobile application to monitor the garbage bins provide information on the amount of rubbish collected. For this they used ultrasonic sensors to measure rubbish level and compare it to the depth of the garbage containers. When the garbage reaches a pre-defined threshold, the system triggers an alert. This alert is transmitted to a central server or directly to municipal authorities via a connected GSM or WiFi module,

ensuring real-time notification about the status of the dustbin. For this a microcontroller platform Arduino Uno is used which processed the data from the sensors and manages communication with external links. This smart dustbin solution not only enhanced operational efficiency but also encouraged better public hygiene and environmental sustainability in rapidly urbanizing areas [6].

Khang Lam et al. [7] constructed a smart domestic waste classification system which can classify the garbages into recyclable and non-recyclable wastes. The system was designed to address the growing concern of household waste management and to promote recycling by automating the sorting process. They trained the module with the images of waste from internet as well as with manually created data, which helped the model learn from a diverse range of waste types and improve its generalization capability across different environments. To further enhance the system's robustness, the team implemented data augmentation techniques and cross-validation during the training phase. They used PIR sensor to detect trash and Raspberry Pi for segregating waste into different bins, a setup that enables the system to operate autonomously with minimal human intervention. The Raspberry Pi, acting as the central processing unit, runs customized algorithms to categorize waste based on real-time image analysis and sensor input. An IR sensor was used to measure volume of trash bin which send notification to user via mobile application if the bin is full, ensuring timely disposal and reducing overflow incidents. The mobile application also provides users with insights into their waste generation patterns and encourages sustainable habits. They achieved an accuracy of 90%, demonstrating the feasibility and effectiveness of integrating artificial intelligence and IoT technologies for practical waste management solutions in households.

Abdullah Alourani et al. [8] addressed the challenges of municipal solid wastes by introducing an intelligent and smart solid waste management system (ISSWMs) to smartly collect and segregate solid waste. The system integrates IoT-enabled smart bins equipped with ultrasonic fill-level sensors and weight sensors to monitor waste accumulation in real time. They collected waste using smart bins connected through mobile applications, which allow users to locate nearby bins, receive status updates, and track disposal metrics, fostering community participation. For automated segregation, the team developed a waste classification model trained using the VGG-

19 deep learning architecture, a pre-trained convolutional neural network (CNN) optimized through transfer learning on a dataset of 15,000 labeled waste images. This model segregates waste into plastic, glass, and metal types, leveraging feature extraction capabilities to identify material textures and shapes with high precision. They achieved a maximum accuracy of 99.7%, a significant improvement over traditional computer vision approaches that often struggle with overlapping waste items or variable lighting conditions. The system also incorporates a cloud-based dashboard for municipalities to analyze waste generation patterns and optimize collection schedules. During field trials, the ISSWMS reduced contamination in recycling streams by 68% compared to manual sorting methods.

The project titled” ConvoWaste: An Automatic Waste Segregation Machine Using Deep Learning” by Md. Shahariar Nafiz, Abdullah Al. Juabir, Shuvra Smaran Das, Dip Nandi and Md. Kishor Morol applied deep learning an image processing techniques to classify the waste precisely and place the detected waste inside the corresponding bins with the help of a servo motor-based system. They classified waste into six categories which ae plastic, organic waste, medical waste, metal, glass and e-waste. They controlled the entire system remotely through an android app in order to dump the separated waste in a desired place by its automation properties. They achieved accuracy of 96% for organic, 94% for plastic and metals, 92% for medical wastes, 90% for glass and 88% for e-wastes [9].

Another similar project titled “A Deep Learning-Based approach to Segregate Solid Waste Generated in Residential Areas” by Sathiyapoobalan Sundaralingam and Neela Ramanathan developed a household waste segregator which classified waste into five categories including paper, plastic, organic, glass and metal. The model was trained using tensorflow to identify the waste category and transmit the data to Arduino which used infrared sensors and motors to control and modify the platform. After matching the platform, the waste was dropped into the dustbin. The model was found to have a mean average precision of 86.5% and average recall of 88% [10].

The project titled “Classification of Waste Materials using CNN Based on Transfer Learning” by Sujana Poudel and Prakash Poudyal performed a comparative analysis on several pre-trained models like InceptionV3, InceptionResNetV2, Xception, VGG19,

MobileNet, ResNet50, and DenseNet201 model used for waste classification. For this, they categorized waste into seven categories as cardboard, glass, metal, organic, paper, plastic, and trash. Among this cardboard, organic and paper were considered as biodegradable and rest as non-biodegradable. They used a total of 3292 images of different types of waste for both training and validation. They preprocessed the images by resizing it and randomly generated additional images from small groups of images by applying transformations like flipping and rotation. Then feature extraction was done by feeding the convolution layer of pre-trained network with new data, and then trained a new classifier on top of it. Fine tuning of the image data was performed before construction of model by training. After the model was trained, performance of these trained models with the help of training accuracy, validation accuracy, and confusion matrix were analyzed. Among them, it was found that ResNet50, InceptionResNetV2, and DenseNet201 models were the top three performing models with accuracy of 98.49%, 98.26% and 98.65% training accuracy and 93.71%, 93.82% and 95.05% validation accuracy respectively. The InceptionV3 model got higher training accuracy in comparison with the remaining models; however, it seemed quite overfitting as it had a high variation between validation and training accuracy. However, the InceptionResNetV2, ResNet50, and DenseNet201 models had less variation between training and validation accuracy [11].

Nine research studies collectively explore advancements in AI- and IoT-powered smart bins and waste segregation systems. They cover the design and deployment of automated waste segregation using machine learning and deep learning models, such as CNNs and transfer learning, for accurate classification of various waste types. Several works highlight the integration of IoT sensors and real-time notifications to optimize waste collection and management. Comparative studies assess the effectiveness of different AI algorithms and sensor systems for smart bins, while some focus on sustainable solutions in specific regions, like Nigeria and South Asia. Overall, these studies demonstrate significant progress in automating and improving waste management efficiency through intelligent technologies.

2.1 EXISTING SYSTEMS

- **Bigbelly Smart bins**

It is a globally recognized smart waste and recycling bin system that integrates solar-powered compactors, sensors, and IoT technology to optimize urban waste management. Each bin is equipped with a built-in solar-powered compactor, allowing it to hold up to five times more waste than a traditional bin by compressing the trash inside. It also features fill-level sensors that monitor the amount of waste inside and transmit real-time data to a cloud-based platform enabling municipalities and facility managers to receive alerts when bins are nearly full. This results in lower operational costs, reduced emissions, and cleaner public spaces.

- **Bin-e**

It is an AI-based smart waste bin designed for use in public places and offices that automatically identifies the waste and sorts them into categories such as plastic, paper, glass, and metal. The input door opens automatically when the user approach it which provides a hands-free operation. It can also compress the plastic and paper wastes automatically to decrease the volume and reduce the frequency of emptying the bin. The integrated app can deliver real-time data including detailed statistics of waste collected and notifying to empty the bin when it is full. It has an accuracy of 92%, making it a reliable system.

- **TrashBot**

TrashBot developed by CleanRobotics is an AI-powered waste-sorting bin designed to make recycling smarter, easier, and more efficient. The camera and sensors scans and identifies the disposed waste and the software decides whether the item is destined for the landfill or recycling facility and sends it into the appropriate bin below. It monitors the fill levels of internal bins and sends alerts when emptying is required, reducing manual checks. It can sort the waste with an accuracy of 96%. The newer models of TrashBot can sort waste into four categories including trash, recycling, compost (food waste), and sometimes hazardous or specialized streams.

- **FoharMalai**

FoharMalai is a Nepali waste management platform designed to address the growing problem of unmanaged solid waste in urban areas of Nepal, particularly in Kathmandu Valley. The system acts as a digital mediator between waste

producers—such as households, offices, and commercial businesses—and authorized waste collectors or recycling companies. Once its smart bin is filled, user sends the notification using the Mobile app to FohorMalai or waste collecting point. Then the truck driver, drives through the best and short route to pick the waste. At the same time, the waste will also be measured using a measuring device to know the weight of the waste. The driver then drives to the waste plant. At the same time, the waste will be then checked. (If the waste is separated properly or not) After that, the point will be rewarded to the citizen. FoharMalai promotes sustainability by encouraging users to separate biodegradable and non-biodegradable waste and by supporting the circular economy through partnerships with recycling firms. The initiative is helping to modernize waste management in Nepal by making the process more transparent, efficient, and environmentally friendly.

2.2 LIMITATIONS OF EXISTING SYSTEM

Some main limitations of these existing systems are

1. The systems like Big belly smart bins, Bin-e and TrashBot are not available in Nepal and require a very high cost.
2. System like FoharMalai doesn't have any AI capabilities for segregating the waste into different categories.

2.3 SOLUTIONS PROVIDED

Our system introduces an AI-based waste segregation capabilities which is currently not available in the existing waste management solutions in Nepal. By leveraging advanced artificial intelligence, the system automatically classifies incoming waste into five distinct categories: organic, plastic, paper, metal and unknown. In addition to this, our system is equipped with real-time waste level monitoring where the sensors embedded within the bins continuously measure the level of accumulated waste, and this data is instantly relayed to home users and waste management personnel through a user-friendly mobile application TrashGold and a desktop website TrashNet respectively. Additionally, our public place bins generate a QR code after every use to the customers which can be scanned through the mobile application to gain points which can be used for redemption.

CHAPTER 3: METHODOLOGY

3.1 PROCESS MODEL

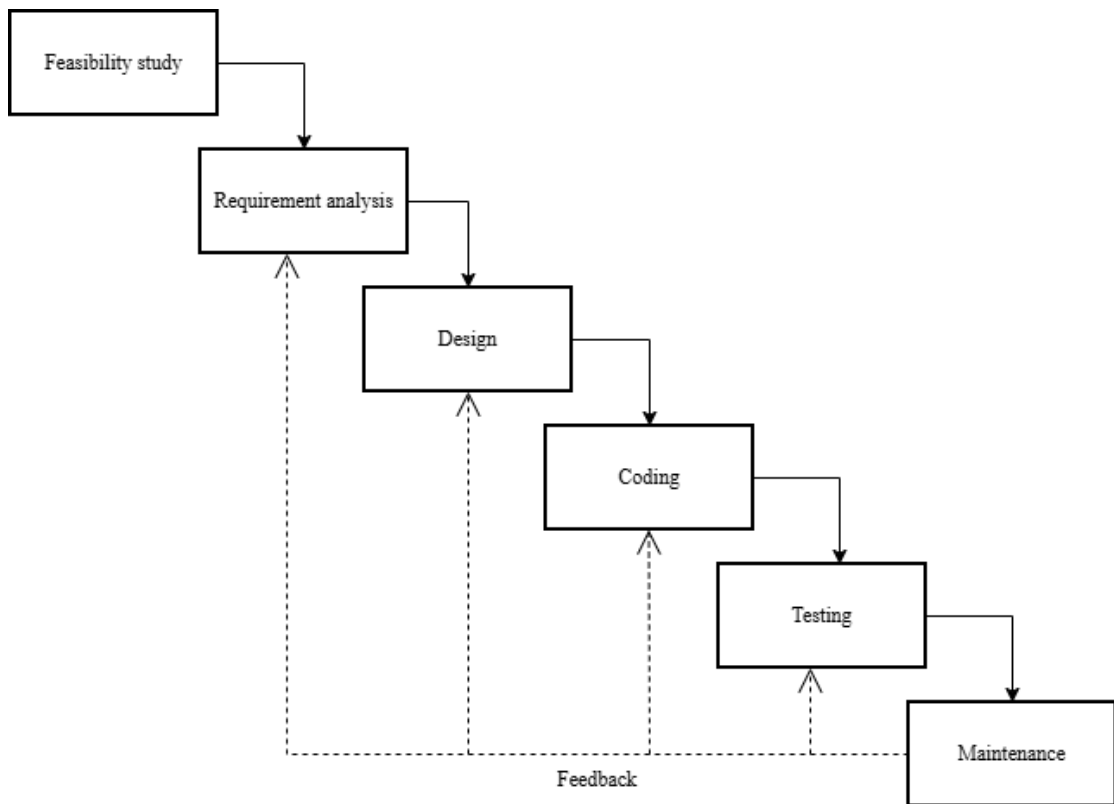


Figure 3: Iterative Waterfall Model

The system is developed using an Iterative Waterfall Model, which provides both structured development phases and flexibility for incremental improvements. This approach offers the following advantages:

Comprehensive Requirement Analysis

The initial phase establishes clear functional requirements including:

- Real-time fill-level monitoring
- Waste classification capabilities
- Cloud-based. data transmission

These specifications is validated with users to ensure technical and operational feasibility. To fulfill above requirements, we use hardware components like ESP32 CAM, Ultrasonic Sensors and Servo Motors.

Iterative Design Refinement

The system architecture undergoes progressive enhancements through:

- Multiple design-review cycles
- Continuous user feedback integration
- Performance optimization at each stage

Particular attention is given to ergonomic factors and user interface design during these iterations.

Modular Development Approach

The implementation follows a component-based methodology:

- Independent development of hardware subsystems
- Separate implementation of machine learning algorithms

This modularity facilitates parallel development and simplify debugging processes.

Phased Verification and Validation

A rigorous testing protocol is implemented:

- Unit testing for all individual components
- Integration testing at subsystem boundaries
- Field testing under operational conditions

This staged verification ensures reliability before full deployment.

System Integration Strategy

The project employs:

- Standardized communication protocols
- Comprehensive interface specifications
- Version-controlled integration milestones

These measures guarantee seamless interoperability between subsystems.

Adaptive Development Framework

The methodology accommodates:

- Requirement evolution
- Emerging technical challenges
- Operational feedback

This flexibility enables continuous improvement while maintaining project timelines.

3.2. SYSTEM BLOCK DIAGRAM

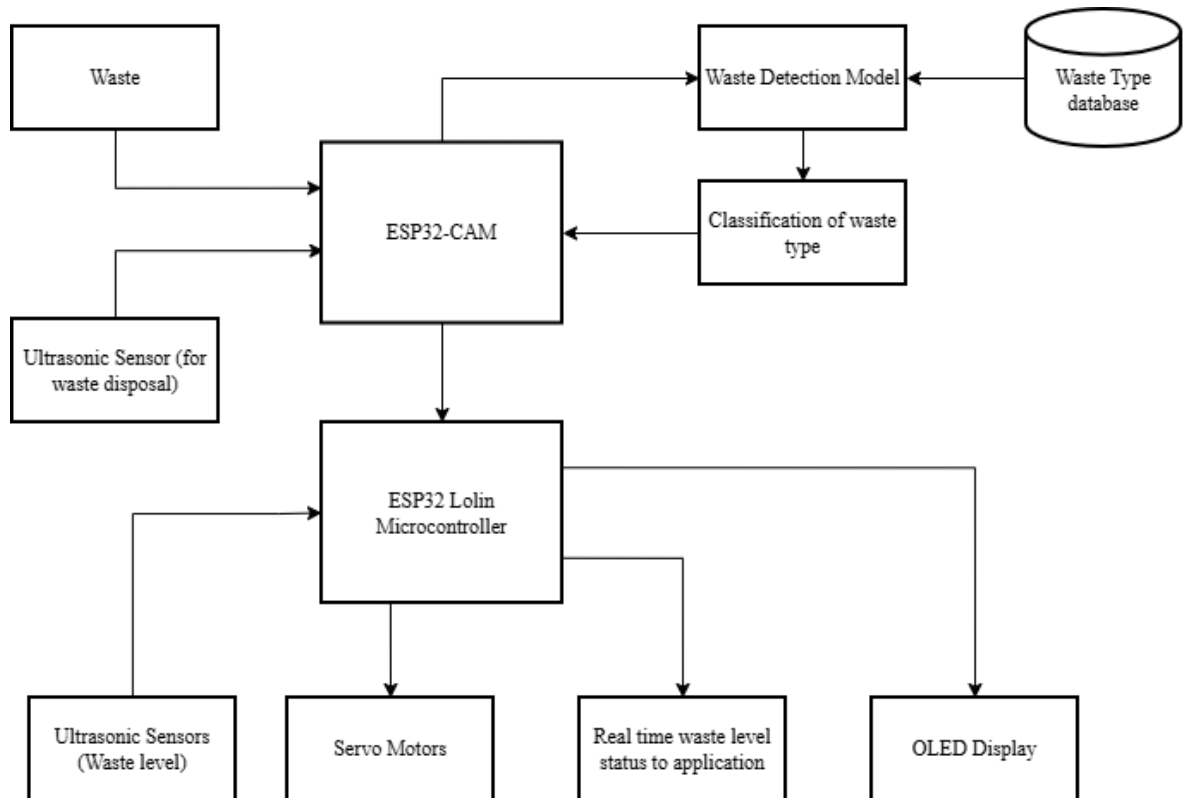


Figure 4: System Block Diagram

Waste: For beginning the disposal process, firstly the user places the waste into the mechanical flap of the smart bin. The waste is placed here before being directed to its appropriate compartment.

Ultrasonic sensor (for waste disposal): An ultrasonic sensor is placed which act as a trigger mechanism for initiating the camera module to capture waste image. After placing the waste onto the flap, the user is required to place their hand near the ultrasonic sensor. The sensor detects the presence of hand by measuring the change in distance between sensor and object. Once the sensor detects the user's hand, it sends a signal to the ESP32-CAM module.

ESP32-CAM: The ESP32-CAM is a camera module serves as a camera unit for capturing the image of the waste placed on the flap. After receiving the trigger signal from the ultrasonic sensor, it activates the camera and captures the image of the waste. Once the image is captured, it is sent to the waste detection module through Wi-fi for

further analysis. Once the classification is completed by the model, the identified waste type is sent back to the system and forwarded to the ESP32 Lolin microcontroller.

ESP32 Lolin Microcontroller: The ESP32 Lolin microcontroller functions as the main controller of the smart bin system. It receives the classified waste type from the ESP32-CAM and determines the appropriate action for waste disposal. Based on the received classification result, the microcontroller controls the servo motors that rotate and tilt the mechanical flap to direct the waste into the correct compartment. In addition to this, the ESP32 Lolin microcontroller also manages the waste level monitoring process. It receives echo time signals from the ultrasonic sensors installed in each compartment of the bin. Upon receiving the echo time, it computes the depth level of the waste bin and send this waste level status to the application for user. If any compartment is 90% full, then it will send alert to the user through the application to inform the user that the bin needs to be emptied.

Waste Detection Model: The waste detection model receives the image frame of waste disposed sent by the camera module and now it classifies the type of waste which includes five categories as: Organic, Plastic, Paper, Metal and Unknown. In the model, comparison of the input waste image is done with the image of labels associated of the waste categories like organic, paper, plastic and metal. The model categorizes the input image to the one with which it has higher confidence. If the input image do not fit to any of these labels, then it is categorized as “Unknown”.

Waste database: This database stores labeled examples of various waste types along with their extracted features. The waste types include organic, plastic, paper and metal. By referencing this dataset, the detection model is able to accurately classify incoming waste images during operation.

Servo Motors: The system uses two servo motors to control the movement of the mechanical flap that directs the waste into proper compartment. The MG996R servo motor is mounted above the MG90S servo motor. Based on the classified waste type, the MG90s servo motor rotates the flap to the proper compartment. And the MG996R servo motor tilts to release the waste to that compartment. Then, both of these servo motors return to theirs default positions.

Ultrasonic sensors (waste level): Each compartment has an ultrasonic sensor for tracking the level of waste in it. The sensor mounted emits high-frequency ultrasonic pulses toward the waste surface which travel through the air, strike the waste, and reflect back to the sensor. The sensor measures the time taken for the echo to return. It sends this information to the ESP32 Lolin microcontroller.

OLED Display: After the waste has been disposed to the proper compartment, the waste type and QR code is displayed by the OLED display which can be scanned using the mobile application to gain points. Each successful disposal in public dustbin increases the user's points, and accumulated points can later be redeemed.

Real time waste level status to application: Through the mobile application TrashGold, the user can view the waste level status of their home smart bin in real time. If the smart bin is 90% full, an alert is sent to prompt the user to empty the bin. For public place dustbins, the TrashNet website shows status of all the dustbins, their locations and analytics.

3.3 FLOWCHARTS

3.3.1 FLOWCHART OF WASTE DISPOSAL PROCESS

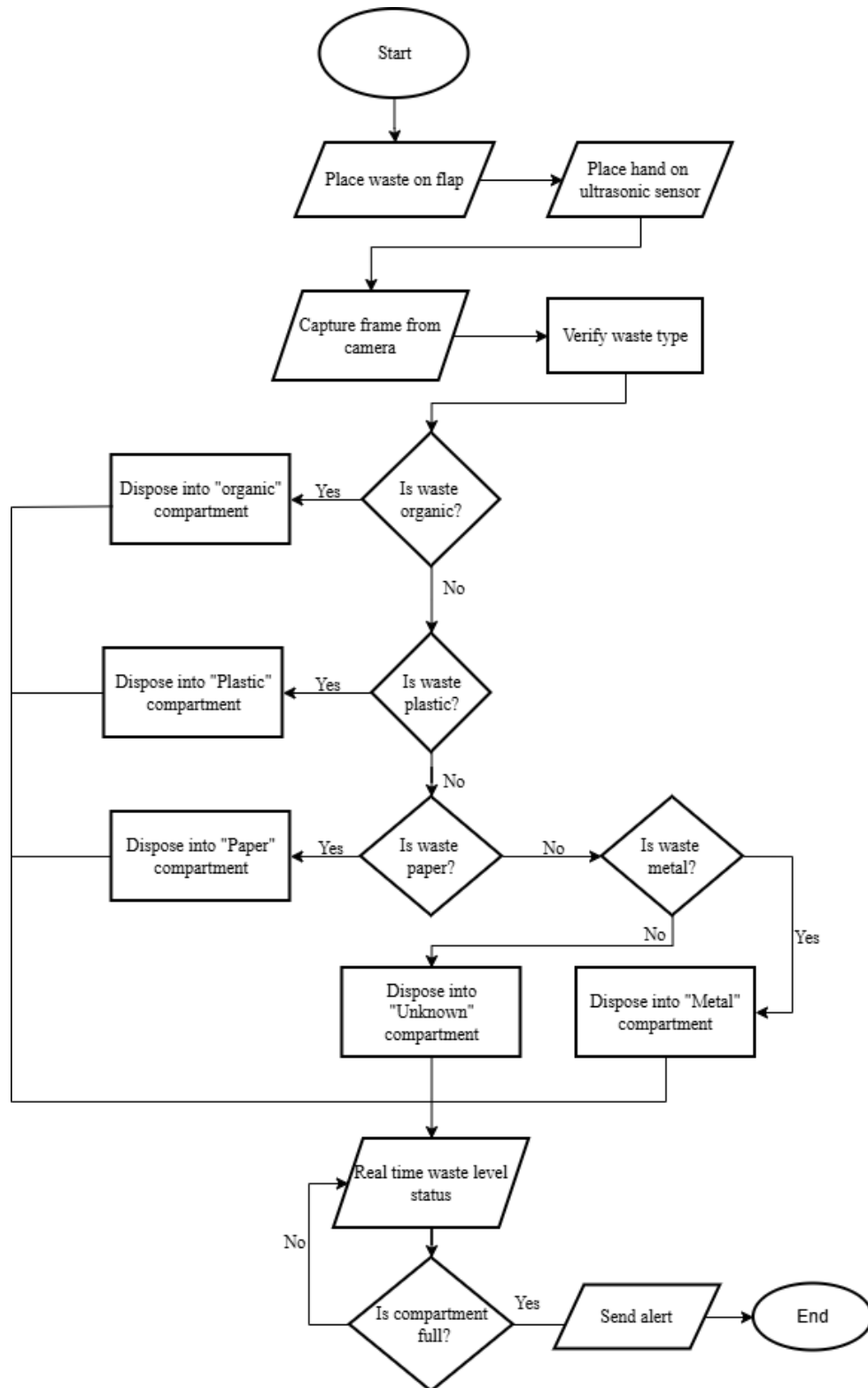


Figure 5: Flowchart of Waste disposal process in the bin

3.3.2 FLOWCHART OF MOBILE APPLICATION TRASHGOLD

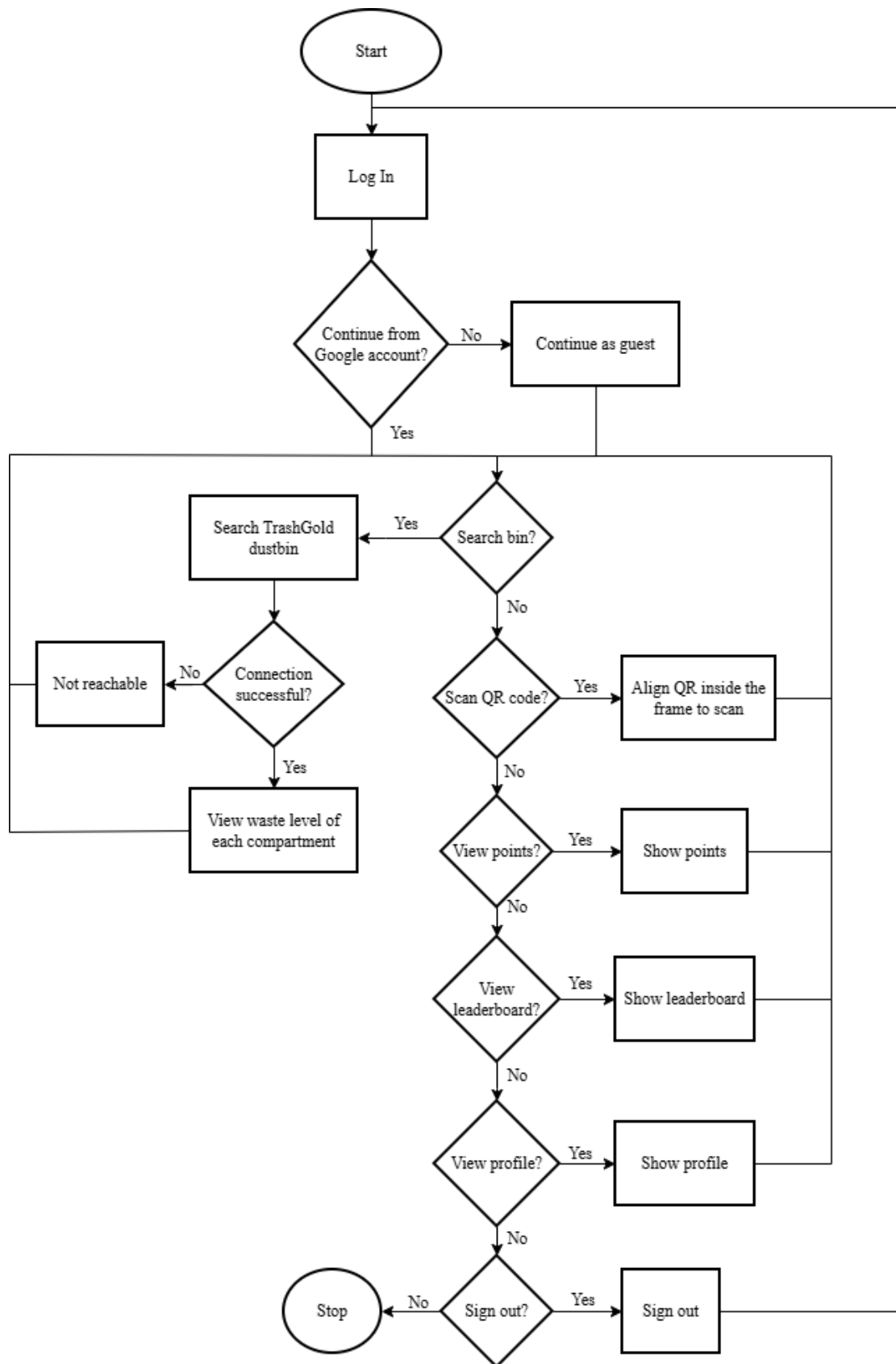


Figure 6: Flowchart of mobile application TrashGold

3.3.3 FLOWCHART OF DESKTOP WEBSITE TRASHNET

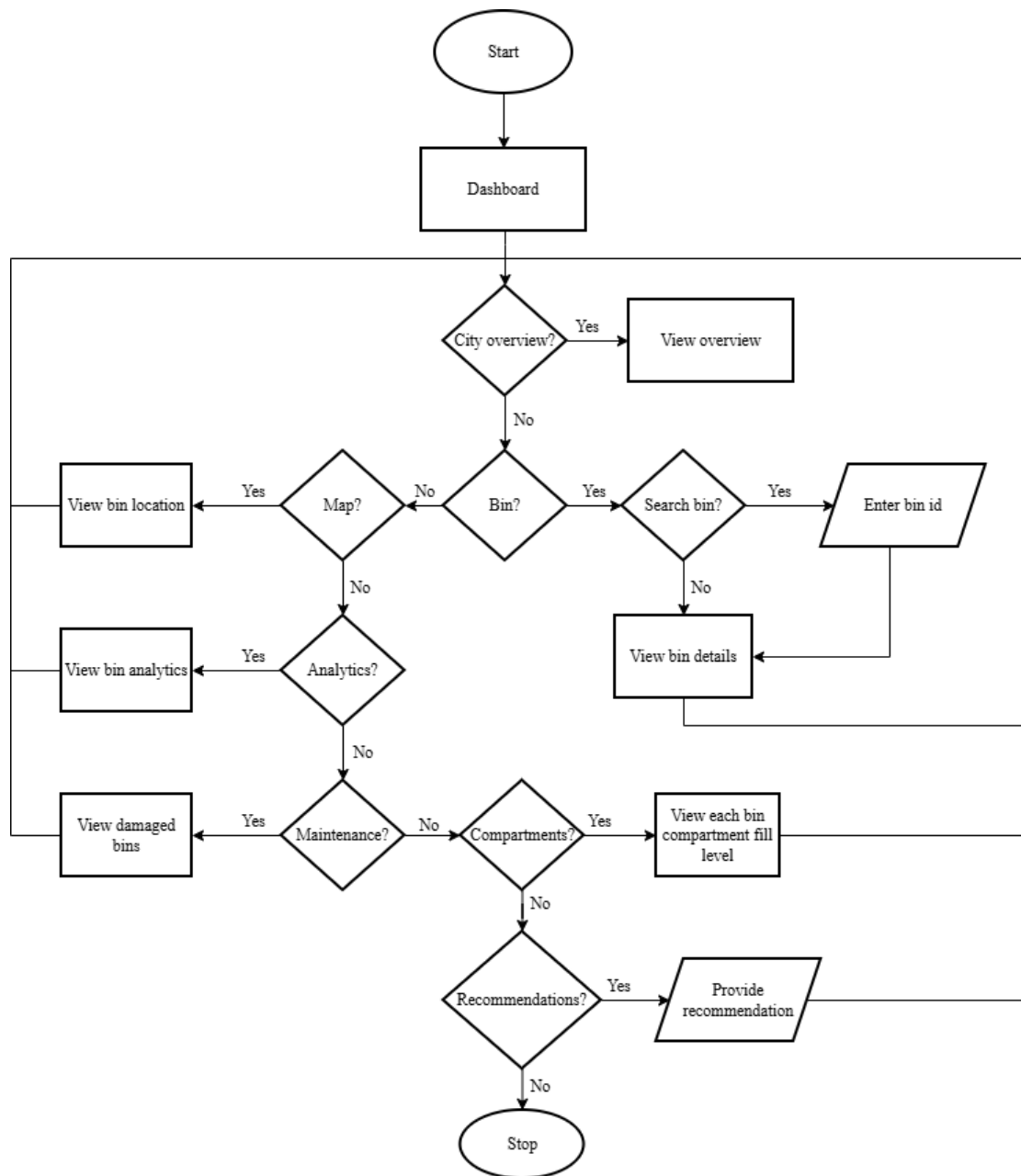


Figure 7: Flowchart of desktop website TrashNet

3.4 ALGORITHM

3.4.1 CONVOLUTIONAL NEURAL NETWORK (CNN)

Convolutional Neural Networks (CNNs) have transformed computer vision by allowing machines to achieve unprecedented accuracy in tasks like image classification, object detection, and segmentation. CNNs, which originated with Yann LeCun's work in the late 1980s, are inspired by the human visual system and process visual data using a hierarchical structure. This article delves into the workings of CNNs, specifically their layers and convolutional operations.

Layers in CNN

The convolutional layer is the primary component of a CNN. It works as a convolution operation by sliding a filter (which is also referred to as a kernel) across the input image and calculating the dot product of the filter and the input's receptive field. This operation aids in detecting local characteristics such as edges, textures, and trends. The layers in CNN are:

i. Convolutional layer

This is the core building block of a CNN. It applies filters (kernels) to the input image to detect various features such as edges, textures, and patterns. Each filter moves across the input, performing element-wise multiplication and summing the results to produce a feature map. Deeper convolutional layers detect more complex features by building upon the patterns detected in previous layers.

Suppose we have an input image described by tensor I of dimension $m_1 \times m_2 \times m_c$, where

m_1 =height of image

m_2 =width of image

m_3 =number of channels

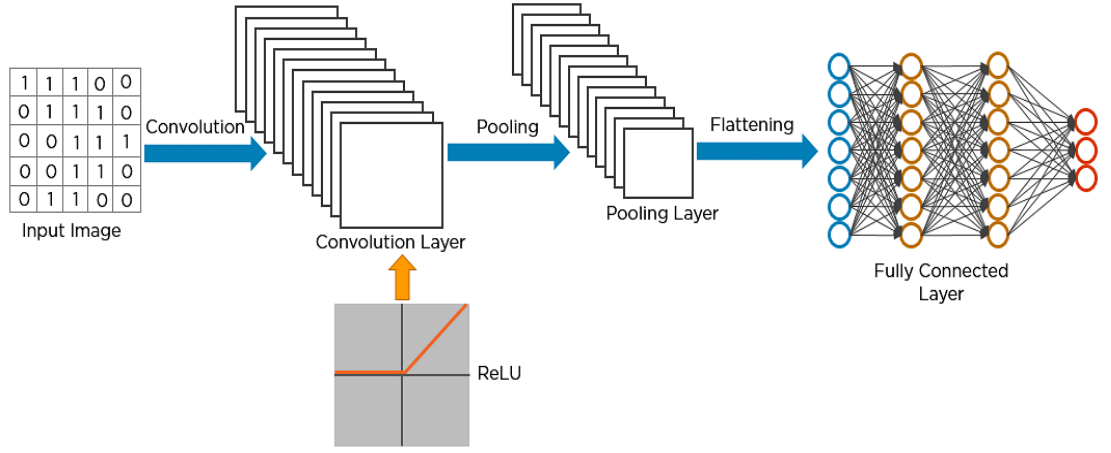


Figure 8: CNN model [12]

We apply a filter which is also a tensor of dimension $(n_1 \times n_2 \times n_c)$. (The number of channels for the kernel is the same as the input image). The filter moves upon the image from left to right, do multiplication between that part of I and K and sum up those products. Stride determines the step size by which the filter would move upon image. The resultant of I and K is another tensor of dimension $(m_1 - n_1 + 1) \times (m_2 - n_2 + 1) \times 1$.

Here,

$$\text{Dim of } I = m_1 \times m_2 \times m_c$$

$$\text{Dim of } K = n_1 \times n_2 \times n_c$$

$$\text{Dim of } F = (m_1 - n_1 + 1) \times (m_2 - n_2 + 1) \times 1$$

And,

$$f[i, j] = (I * K)_{[i, j]}$$

The ij -th entry of the feature map is given as below:

$$f[I, j] = \sum_x^{m_1} \sum_y^{m_2} \sum_z^{m_c} K_{[x, y, z]} I_{[i+x-1, j+y-1, z]}$$

We have taken the following example of a $5 \times 5 \times 1$ dimensional image being convoluted with a kernel of $3 \times 3 \times 1$ and the stride $s = 1$ has been used.

1	0	1	1	0
1	0	1	0	0
1	0	1	0	1
0	1	0	0	1
1	0	0	1	1

Image

*

1	0	1
0	1	0
0	0	1

Kernel

=

4	2	3
2	2	2
4	1	3

Feature map

The ij-th entry of feature map is given by following general formula in case of single channel:

$$f[i, j] = (I * K)_{[i,j]} = \sum_{x=1}^{m_1} \sum_{y=1}^{m_2} K_{[x,y]} I_{[i-x, j-y]}$$

Let us compute the 11-entry of the feature map in above example:

$$f[i, j] = \sum_{x=1}^{m_1} \sum_{y=1}^{m_2} \sum_{z=1}^{m_c} K_{[x,y,z]} I_{[i+x-1, j+y-1]}$$

$$\begin{aligned} f[1, 1] &= \sum_{x=-2}^5 \sum_{y=-2}^5 K_{[x,y]} I_{[1-x, 1-y]} \\ &= \sum_{x=-2}^5 K_{[x,-2]} I_{[1-x, 3]} + \sum_{x=-2}^5 K_{[x,-1]} I_{[1-x, 2]} + \\ &\quad \sum_{x=-2}^5 K_{[x,0]} I_{[1-x, 1]} + \sum_{x=-2}^5 K_{[x,1]} I_{[1-x, 0]} + \sum_{x=-2}^5 K_{[x,2]} I_{[1-x, -1]} \end{aligned}$$

$$\begin{aligned} f[1, 1] &= K[-2, -2] I[3, 3] + K[-1, -2] I[2, 3] + K[0, -2] I[1, 3] + K[1, -2] I[0, 3] + K[2, -2] I[-1, 3] + K[-2, -1] I[3, 2] + K[-1, -1] I[2, 2] + K[0, -1] I[1, 2] + K[1, -1] I[0, 2] + K[2, -1] I[-1, 2] + K[-2, 0] I[3, 1] + K[-1, 0] I[2, 1] + K[0, 0] I[1, 1] + K[1, 0] I[0, 1] + K[2, 0] I[-1, 1] + K[-2, 1] I[3, 0] + K[-1, 1] I[2, 0] + K[0, 1] I[1, 0] + K[1, 1] I[0, 0] + K[2, 1] I[-1, 0] + K[-2, 2] I[3, -1] + K[-1, 2] I[2, -1] + K[-1, 2] I[2, -1] + K[0, 2] I[1, -1] + K[1, 2] I[0, -1] + K[2, 2] I[-1, -1] \end{aligned}$$

The entries which are not available are substituted zero.

$$f[1, 1] = 0 + 0 + 0 + 0 + 0 + 0 + (1 \times 1) + (0 \times 1) + (1 \times 0) + (0 \times 1) + 0 + 0 + (1 \times 0) + 0 + 0 + 0 + 0 + (1 \times 1) + 0 + 0 + (1 \times 1) + 0 + 0 + 0 + 0 + 0 + 0$$

$$f[1, 1] = 3$$

Similarly, the remaining entries can be calculated by using the formula.

ii. Activation function (ReLU)

After convolution, the Rectified Linear Unit (ReLU) activation function is applied to introduce non-linearity. Since real-world data is often non-linear, ReLU ensures

that the model can capture complex relationships by converting all negative values to zero while keeping positive values unchanged.

In general, one bias term 'b' has been added to the convoluted part and then the activation function is applied.

$$c = F + b$$

$$c = I * K + b$$

$$\begin{aligned} \text{Conv}(I, K) &= \varphi_a(c) \\ &= \varphi_a(I * K + b) \end{aligned}$$

Where φ_a is an activation function.

There are different types of activation functions as sigmoid, tangent, hyperbolic tangent function. The most commonly used activation function is ReLU which eliminates the negative values:

$$R(x) = \max(0, x)$$

-25	14	120	ReLU function →	0	14	120
71	38	18		71	38	18
-30	91	-78		0	91	0

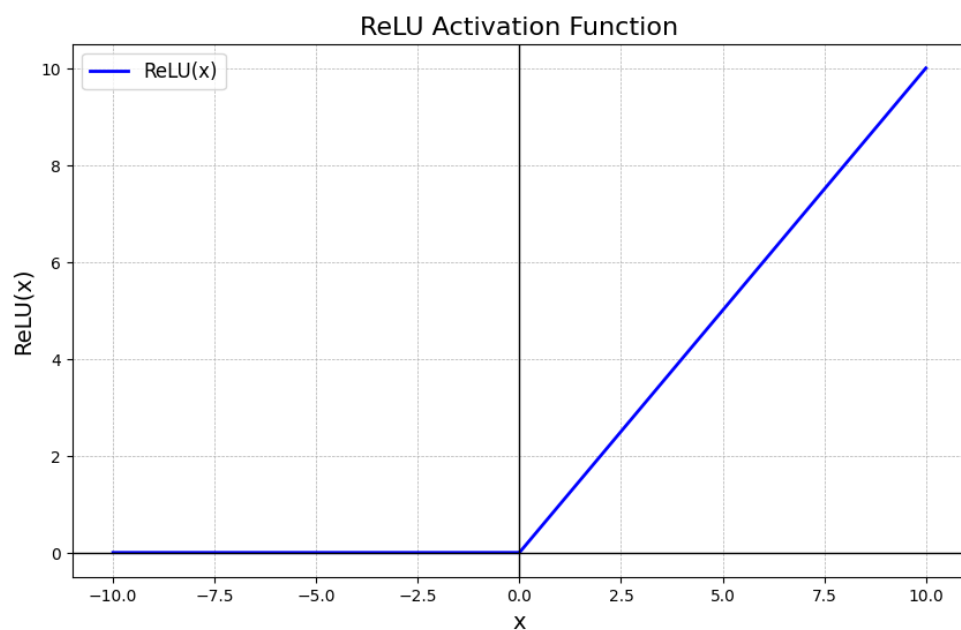


Figure 9: ReLU Function Graph

iii. Pooling layer

Pooling layers reduce the spatial dimensions of feature maps while retaining the most important information. The most common type is max pooling, which selects the maximum value in each region of the feature map. Pooling helps reduce computational load and makes the model more robust to variations in the input, such as minor translations or distortions.

Let us assume that:

$$\text{Conv}(I, K) = C$$

$$P = \phi_p(C)$$

Where ϕ_p is an pooling function.

The dimension of pooled part is given as:

$$\text{dim of } P = \left(\frac{m_1 + 2p - n_1}{s}\right) \times \left(\frac{m_2 + 2p - n_2}{s}\right) \times m_c$$

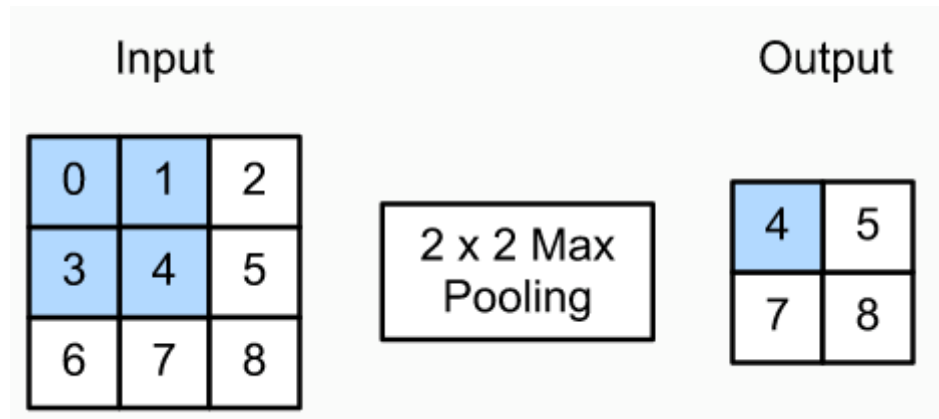
where,

$m_1 \times m_2$ = the dimensions of input image

$n_1 \times n_2$ = the dimensions of padding kernel

here s stands for stride and p stands for padding.

There are different types of pooling such as sum pooling, average pooling, max pooling. Example of max pooling is given below. The maximum pooling is done upon 2×2 patches. From each patch, the maximum entry is selected.



iv. Fully Connected Layer

The final layers in a CNN are fully connected layers, where neurons have connections to all activations from the previous layer. These layers interpret the extracted features and perform classification. The last fully connected layer

typically uses a softmax activation function for multi-class classification, assigning probabilities to different output classes.

Mathematical Formulation

The convolution operation between input image I and kernel K is defined as:

$$(I * K)(i, j) = \sum_m \sum_n I(i + m, j + n) \cdot K(m, n)$$

Where:

- I = Input image (e.g., $224 \times 224 \times 3$ RGB waste image)
- K = Kernel/filter (e.g., 3×3 edge detector)
- (i, j) = Coordinates in the output feature map

Workflow of CNN

Convolutional Neural Networks (CNNs) are designed to process and analyze visual data by learning spatial feature hierarchies automatically and adaptively. Here's a thorough explanation of how CNNs operate:

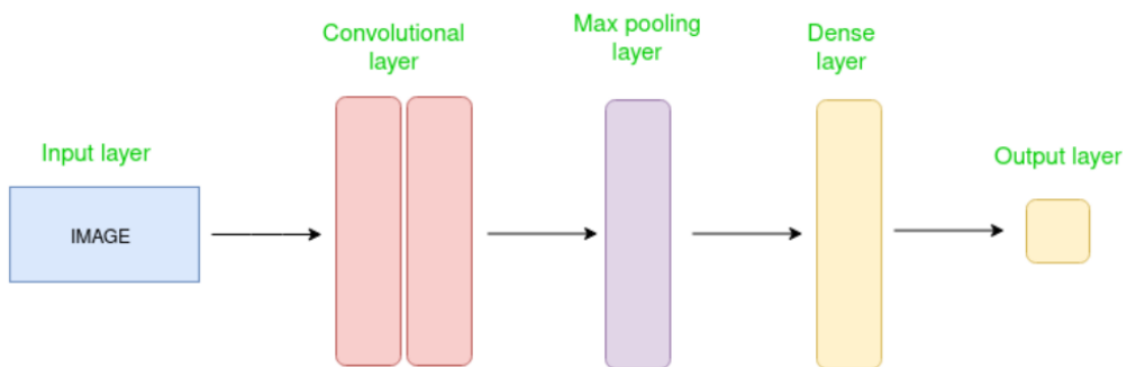


Figure 10: Simplified CNN (Convolutional Neural Network) architecture [13]

Input Layer

Takes in waste images (like a photo from the bin's camera). Color images use 3 layers (red, green, blue), while black-and-white uses 1.

Convolution Layers

Small filters (like 3x3 grids) slide across the image to spot patterns. The first layers find simple things like edges, while deeper layers recognize complex shapes like bottles or food wrappers.

Padding & Stride

Padding adds a border around the image to keep size the same. Stride controls how fast the filter moves - bigger stride means smaller output.

Activation (ReLU)

Turns negative numbers to zero, keeping only important features. Makes the network learn faster.

Pooling

Shrinks the image by keeping only the strongest features in each area (max pooling) or averaging them (average pooling).

Deep Layers

Early layers find simple patterns, later layers combine these to recognize whole objects like "plastic bottle" or "paper cup".

Final Layers

Flattens everything and connects to output neurons that decide the waste type (recyclable, organic etc.).

Dropout

Randomly ignores some neurons during training to prevent memorizing instead of learning.

3.5 UML DIAGRAMS

3.5.1 USE CASE DIAGRAM

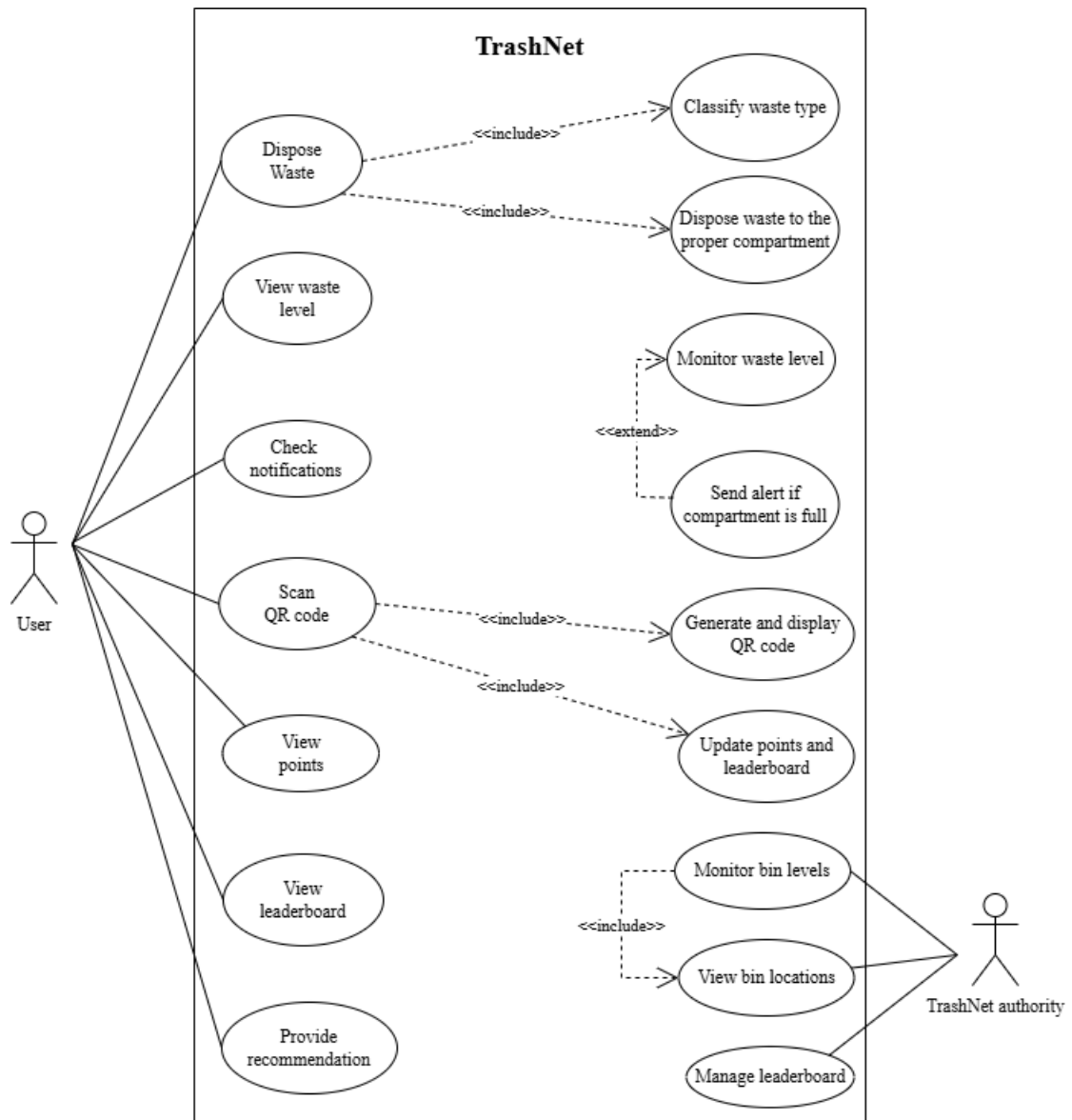


Figure 11: Use Case Diagram of TrashNet

3.5.2 SEQUENCE DIAGRAM

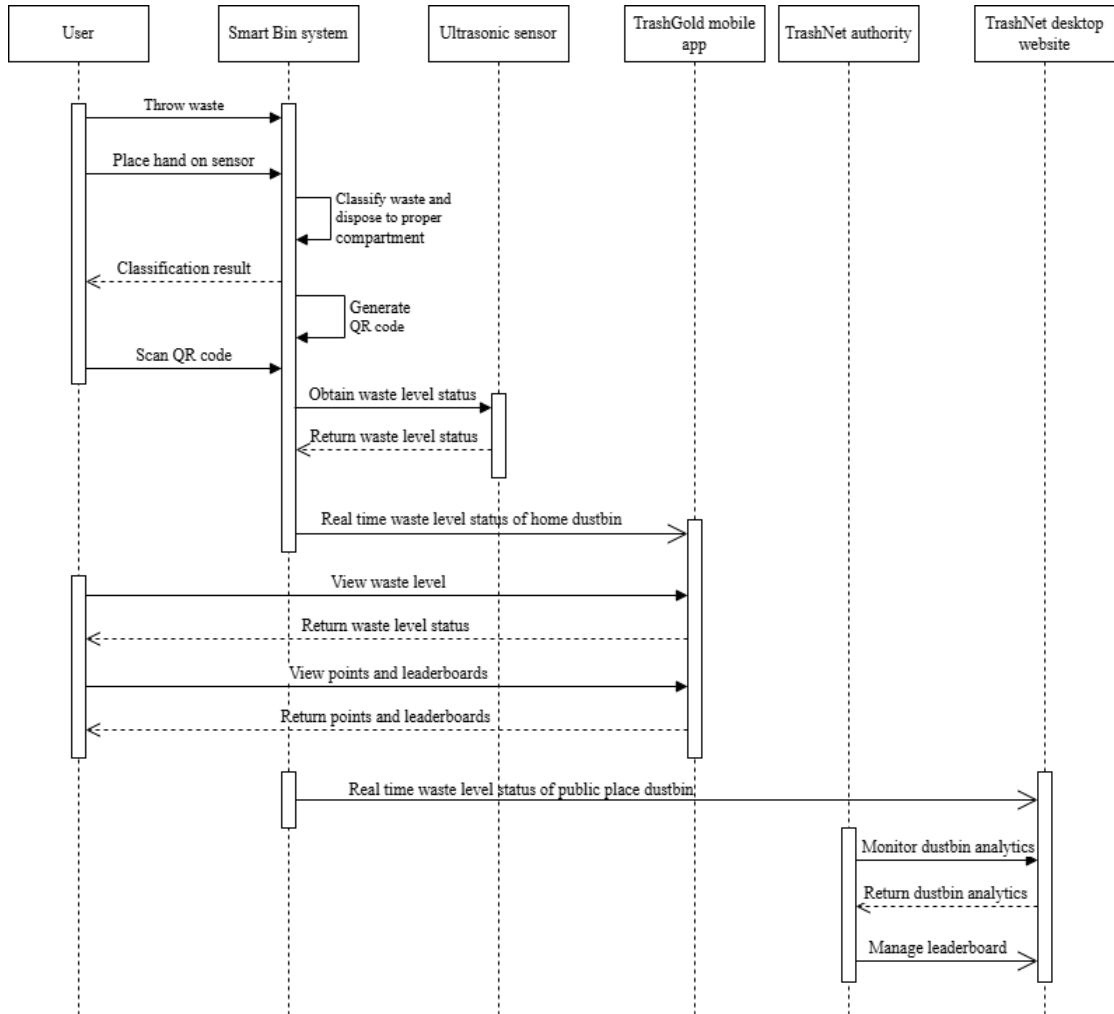


Figure 12: Sequence Diagram of TrashNet

3.5.3 CLASS DIAGRAM

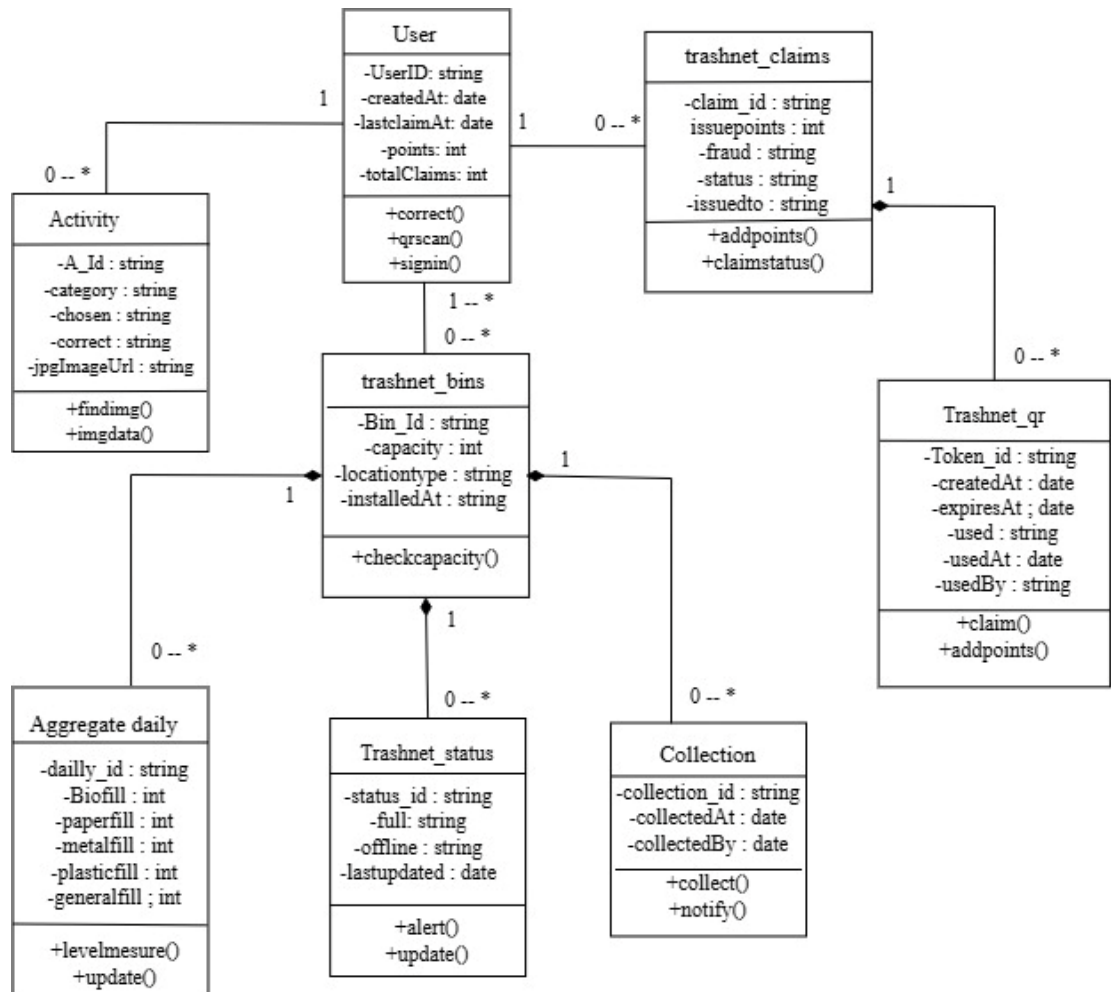


Figure 13: Class Diagram of TrashNet

3.6 VERIFICATION AND VALIDATION

3.6.1 VERIFICATION

Dataset Verification:

- Model can classify images of highly variable physical features within the given label class.
- Model can detect the objects within the image consistently even when positioned at different unconventional angles.
- Dataset for each label class was distinct so that model was trained with domain-appropriate data only.



Figure 14: Sample of Dataset use for AI Model Training

Functional Verification

Core System Functionalities were verified manually under controlled testing environment as:

- Detection accuracy for each label class of detection model was reviewed for the confidence level of each detection.
- Model output was verified to distinguish between objects with similar features that exist on different label class.

Training Process Verification:

The Training Process Verification Graph below showcases that both Training and Validation Loss steadily decreases across epochs for both Bounding Box Loss as well as Class Loss for the overall Model.

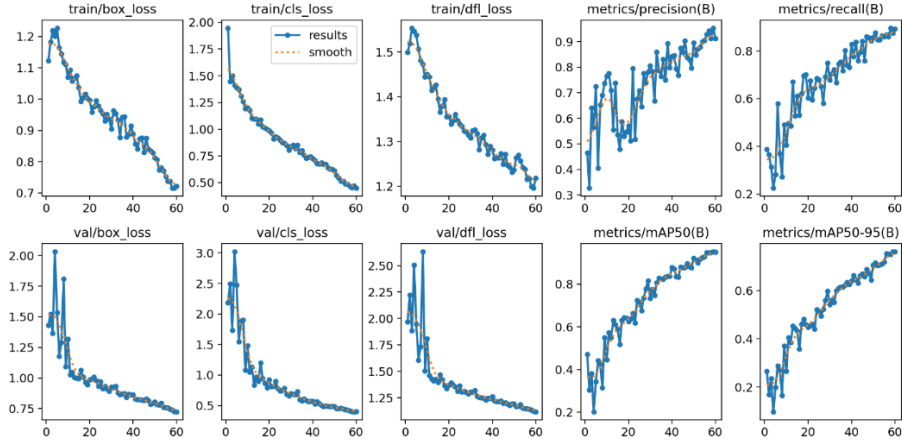


Figure 15: Training and Validation Loss

3.6.2 VALIDATION

Model Evaluation Testing:

For this testing, Train-Test Split method was utilized. For this process, the dataset was split into two parts: Training Dataset and Validation Dataset. The training dataset was annotated and used to train the model and consists of above 500 annotations per class and a total of about 4000 annotations. The Validation Dataset consists of images into trained datasets to determine how well the Model performs within its internal environment. During the testing, Model would draw Bounding Box around the objects that fit under any detection label class. Multiple rounds of Validation were performed to make better use of available dataset.

The Different Class Labels can be recognized as follows:

Label Class	Common Name
0	Biodegradable
1	Metal
2	Paper
3	Plastic
4	Unknown

Table 1: Class labels and names

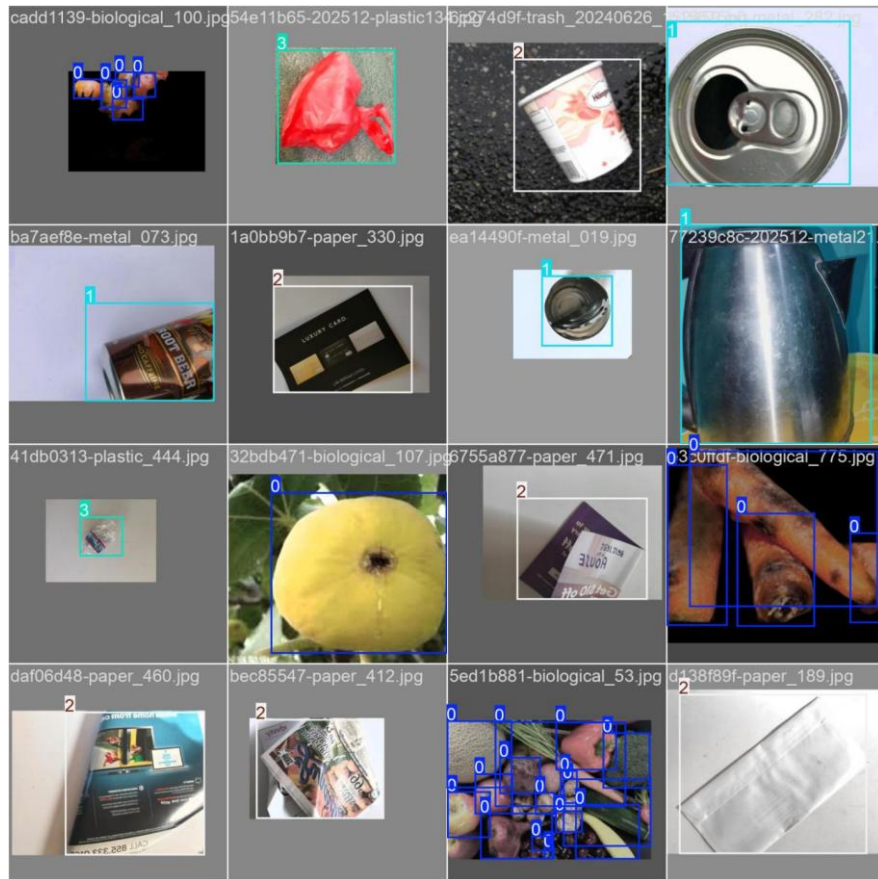


Figure 16: Model Detection Performance During Validation

Human Validation:

The detection model was manually reviewed to assess:

- Detection accuracy for each label class
- Detection accuracy in different forms of media(images, videos, live feed, etc)
- Confidence Level for detection of objects in different angles and lighting conditions.

This assessment confirmed that the detection model is qualified to classify wastes under various conditions with very minor anomalies.

During Validation Testing, certain anomalies were detected. Some of the anomalies along with their potential cause and effect to the model are detailed below:

S.N	Anomaly	Potential Cause	Remark
1.	The model certain Plastic Utensils, especially colored water bottles as Metal and vice versa	Certain colored metal utensils like Bottles looked similar to high quality Plastic Bottles in images due to similar physical features.	More focus on annotating and training Metallic and Plastic Utensils data should help improve Models Detection.
2.	If an image has high frequency objects at once (>50) then the Model is biased towards Biodegradable label class.	In the dataset, Biodegradable objects occur in high frequency (>50) in a single image compared to all other label class.	Very high frequency of items (>50) cannot be put at the same time inside a dustbin. As this problem only occurs with very high number of objects in a single image, this should not be a problem during practical use.
3.	Detection with Live Camera(not Image or Videos), causes model bias towards Paper label class.	In the dataset, Paper Label Class includes Photos of People, Objects, etc. When the Model runs on Live Camera and detects a person's face, it mistakes it for a photo.	The model will always be used to detect object in a image file (.png, .jpg). Hence, in practical use this should not be an issue.
4.	Certain Plastic wrapping is detected as Paper and vice versa.	Various Paper and Plastic wrapping look similar to one another in images.	More focus on annotating and training Plastic Wrapping data should help improve Models detection.

5.	Despite working normally in Internal Testing, the Model showed bias towards Metal and Paper in practical environment.	The cover of bin is Metal. This cover is visible when camera captures photo showing bias towards Metal.	Paper and Metal are given lowest priority for detection in case of multiple objects in an image.
6.	The model only runs when hand is placed close to Ultrasonic Sensor.	Bottleneck due to hardware components of ESP32 and Sensors.	It could be fixed with changing certain hardware components. Not worth pursuing currently due to high cost.

Table 2: Anomalies and their potential causes

3.7 TOOLS USED

3.7.1 SOFTWARE TOOLS

- **Flutter:** Flutter is an open-source UI software development toolkit created by Google for building cross-platform applications from a single codebase. It is primarily used to develop mobile apps for Android and iOS, but it also supports web, desktop, and embedded platforms. Flutter uses the Dart programming language and provides a rich set of customizable widgets to create attractive and responsive user interfaces. It uses its own high-performance rendering engine to ensure smooth animations and consistent design across different devices. With strong community support and continuous updates, Flutter is increasingly adopted for developing modern, scalable applications.
- **Python:** Python is a high-level, interpreted computer programming language that is often used for different tasks like building websites and software, artificial intelligence, tasks automation, data analysis, and scientific computing. It supports multiple programming paradigms, including object-oriented and functional programming. With its large and active community contributing to its development and maintenance, it has many frameworks and libraries which can be used for a variety of purposes.
- **OpenCV:** OpenCV (Open-Source Computer Vision Library) is an open-source library for computer vision, machine learning and image processing. It is developed by Intel and now maintained by a community of developers under the OpenCV foundation. It plays a major role in real-time operation important to today's systems and provide easy-to-use computer vision and image processing capabilities like image stitching, feature detection, video stream processing, camera calibration and diverse image pre-processing tasks.
- **TensorFlow:** TensorFlow is an open-source machine learning and deep learning framework developed by Google. It is widely used for building and training artificial intelligence models, including neural networks for tasks such as image recognition, natural language processing, and predictive analysis. TensorFlow provides flexible tools and libraries that allow developers to design, train, and deploy models efficiently across different platforms, including mobile devices,

servers, and cloud systems. It supports both high-level APIs for beginners and advanced customization for researchers, making it suitable for a wide range of AI applications.

- **C++:** C++ is a powerful, high-performance programming language developed as an extension of the C language. It supports multiple programming paradigms, including procedural, object-oriented, and generic programming, making it highly versatile. C++ allows to write efficient code with direct control over memory and hardware, which is why it is widely used in system programming, embedded systems, and real-time applications like ESP-32 microcontrollers. It provides features such as classes, inheritance, polymorphism, templates, and the Standard Template Library (STL), enabling structured and reusable code.
- **HTML:** HTML (Hyper Text Markup Language) is the standard language used to create and structure content on the web. HTML provides a structured way to organize information so that web browsers can interpret and display content correctly. It is simple to learn and widely supported across all modern browsers. Because of its fundamental role in the World Wide Web, HTML is considered the backbone of all websites.
- **Firebase:** Firebase is a cloud-based platform developed by Google that provides a complete backend infrastructure for application development. It offers a set of integrated services including authentication, databases, cloud storage, analytics, hosting, and serverless functions, all accessible through unified APIs and SDKs. It follows a Backend-as-a-Service (BaaS) model, eliminating the need for manual server management while enabling real-time data synchronization, secure access control through rules, and automatic scalability. It supports NoSQL databases, integrates seamlessly with web and mobile environments, and includes built-in monitoring and performance tools to maintain reliability and efficiency.

3.7.2 HARDWARE TOOLS

- **ESP32-CAM:** The ESP32-CAM is a compact and low-cost camera module that integrates an ESP32 microcontroller, Wi-Fi, and Bluetooth connectivity on a single board. It features the OV2640 camera sensor, which supports image resolutions up to 1600×1200 (UXGA) and provides good image quality for embedded vision tasks. The ESP32 chip includes a dual-core processor, built-in Wi-Fi (802.11 b/g/n),

and Bluetooth (Classic and BLE), enabling wireless image transmission and remote control. Due to its small size, low power consumption, and integrated wireless capabilities, the ESP32-CAM is widely used in AI-based vision systems, IoT applications, and real-time image monitoring solutions.

- **ESP32-Lolin:** The ESP32 Lolin is a development board based on the ESP32 microcontroller, designed to simplify the development of IoT and embedded systems applications. It features built-in Wi-Fi and Bluetooth (Classic and BLE), a dual-core processor running up to 240 MHz, and multiple GPIO pins supporting interfaces such as ADC, DAC, PWM, SPI, I2C, and UART. The Lolin32 board includes a USB-to-serial converter, micro USB port, voltage regulator, reset and boot buttons, and often a Li-Po battery connector, making it convenient for rapid prototyping.
- **Ultrasonic Sensor:** An ultrasonic sensor is an instrument that measures the distance to an object using ultrasonic sound waves. The sensor consists of two primary components: a transmitter and a receiver. The transmitter is responsible for emitting a high-frequency sound. In essence, ultrasonic refers to frequencies beyond the range of the human hearing - so something higher than 20kHz.
- **OLED Display:** An OLED display (I2C) is a small, low-power display module that uses OLED (Organic Light Emitting Diode) technology and communicates through the I2C (Inter-Integrated Circuit) protocol. These displays do not require a backlight because each pixel emits its own light, resulting in high contrast, low power consumption, and clear visibility even in low light. The I2C interface uses only two communication lines—SDA (data) and SCL (clock)—along with power (VCC) and ground (GND), making it ideal for microcontrollers like the ESP32. Common OLED I2C modules (such as 0.96-inch 128×64 displays) are widely used in IoT projects, embedded systems, sensor monitoring, and portable devices due to their compact size and easy wiring.
- **MG996R Servo Motor:** The MG996R is a high-torque digital servo motor commonly used in robotics, RC vehicles, robotic arms, and automation projects. It operates on a 4.8V–7.2V power supply and provides high torque (around 9–11 kg·cm at 6V), making it suitable for heavier mechanical loads compared to smaller servos like SG90. The MG996R uses PWM (Pulse Width Modulation) signals for position control, typically allowing rotation from 0° to 180°. It features metal gears

for improved durability and accuracy, though it consumes relatively high current and often requires an external power source when used with microcontrollers such as the ESP32.

- **MG90S Servo Motor:** The MG90S servo motor is a compact and lightweight motor operating on a 4.8–6 V DC supply. It provides a torque of approximately 1.8–2.2 kg·cm with a rotational range of about 180°. The motor is controlled using PWM signals, enabling precise angular positioning. Its fast response and smooth motion make it reliable for accurate movement control.

3.8 DATASET COLLECTION

The data used in the project was primarily collected from the Garbage Dataset by Suman Kunwar and Garbage classification database of Mohammed Mostafaaba, both of which are available on Kaggle, which forms the majority of the dataset. All the data in these datasets are in image format and consist of garbage of different classes. In addition to this, extra data was also collected manually by capturing images of household and domestic waste.

CHAPTER 4: EPILOGUE

4.1 RESULT

The model is detecting trained objects and successfully distinguishing between different label classes. As seen in the normalized Confusion Matrix, all have high True Positive ratio with the worst False Positive occurring with Biodegradable Class and Background.

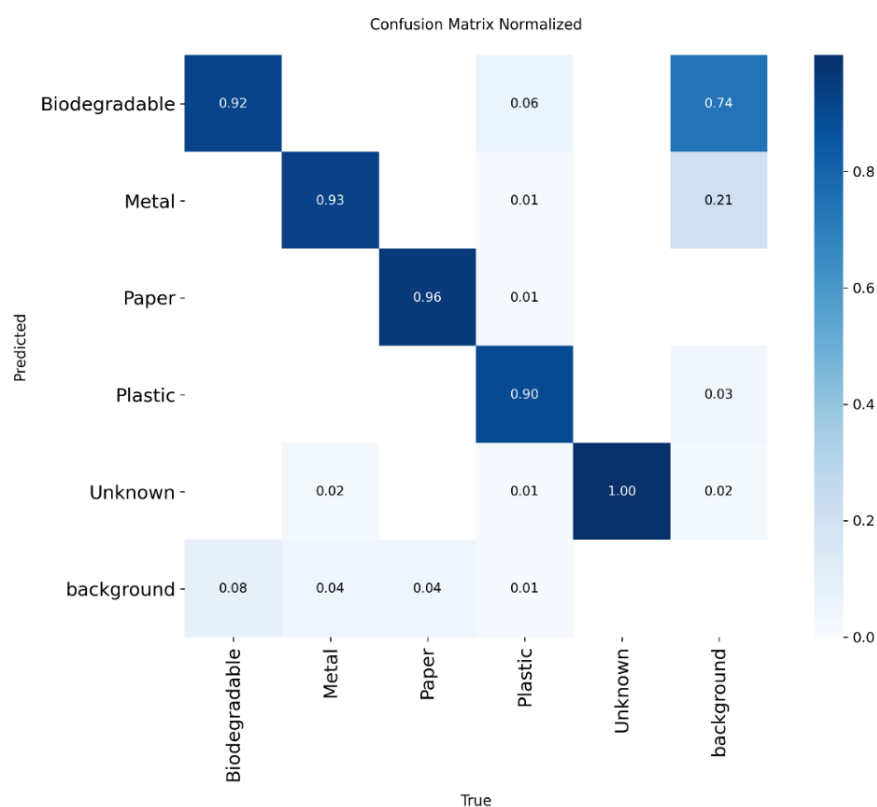


Figure 17: Confusion Matrix of the Model

The F1-Confidence Curve below shows high peak confidence value for each class with the worst being with Biodegradable class. The curve shows 91% accuracy for model with a 61% confidence.

The F1-Confidence Score is calculated as:

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Where,

Precision is the measure of relevant retrieved predictions based on all retrieved detections. It is given by:

$$\text{Precision} = \frac{TP}{TP+FP}$$

Recall is the measure of relevant retrieved predictions based on all relevant instances.

It is given by:

$$\text{Recall} = \frac{TP}{TP+FN}$$

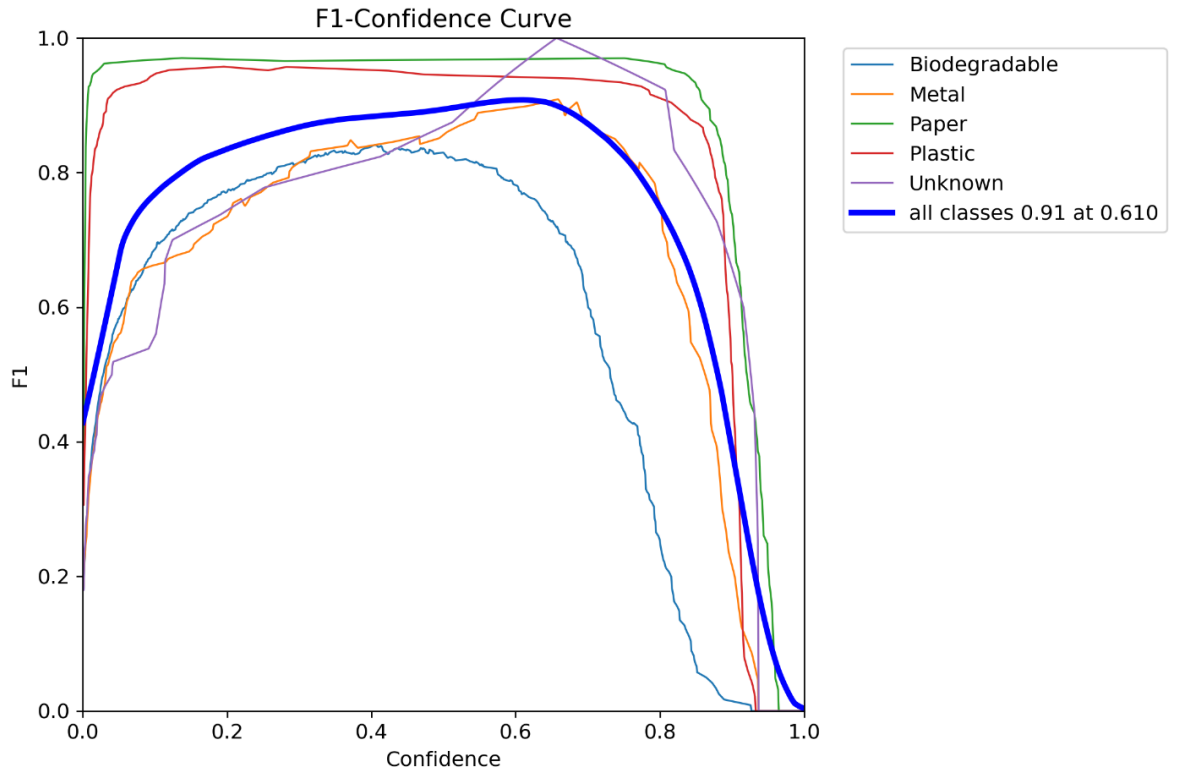


Figure 18: F1-Confidence Curve of the Model

4.2 CONCLUSION

The system successfully addresses the limitations of conventional bins by using smart bins to automatically segregate and monitor the waste level in real time. Using image-based classification and controllers, it classifies the waste into organic, plastic, paper and metal types and segregates them into separate compartments of the bin. Furthermore, sensor-based level detection system ensures accurate and real-time monitoring of waste levels in each compartment of the smart bin.

The system is also integrated with both a mobile application TrashGold and a desktop website TrashNet to enable efficient monitoring and management of waste. TrashGold

application is designed for individual home users, allowing them to view the real-time fill level of each compartment and receive alert when the bin reaches a critical level, such as 90% capacity. TrashNet website is primarily used for public waste management, where authorities can monitor multiple dustbins simultaneously. It provides detailed analytics information such as bin location, compartment-wise fill levels, and collection history, enabling better planning and resource allocation.

In addition, the project promotes user engagement and responsible waste disposal through a QR code-based redemption system and leaderboard features. This not only improves waste management efficiency but also encourages sustainable habits among users. Overall, the system demonstrates a scalable and intelligent solution for modern waste management, combining automation, IoT, and user interaction to create a cleaner and more organized environment.

4.3 FUTURE ENHANCEMENT

1. On device waste classification

Rather than using an external server for running the waste classification model, it could be run on the hardware of the bin itself. This eliminates the dependency on external server and ensures that the system can operate without an internet connection reducing latency and improving response time.

2. Improved AI accuracy

Performance of the bin can be enhanced by improving the accuracy of the AI model used for waste classification. This can be achieved by training the model with a larger and more diverse dataset that includes different types, shapes, and conditions of waste. Higher accuracy will reduce misclassification and ensure more effective segregation, leading to better recycling outcomes.

3. Upgrading to high-end hardware

The system performance and durability can be improved by replacing low-cost components with high-end hardware. Using advanced sensors, more powerful microcontrollers, and robust mechanical components can enhance accuracy, speed, and reliability of waste segregation. High-quality hardware also ensures longer lifespan, better resistance to environmental conditions, and reduced maintenance requirements.

4. **Mass Implementation**

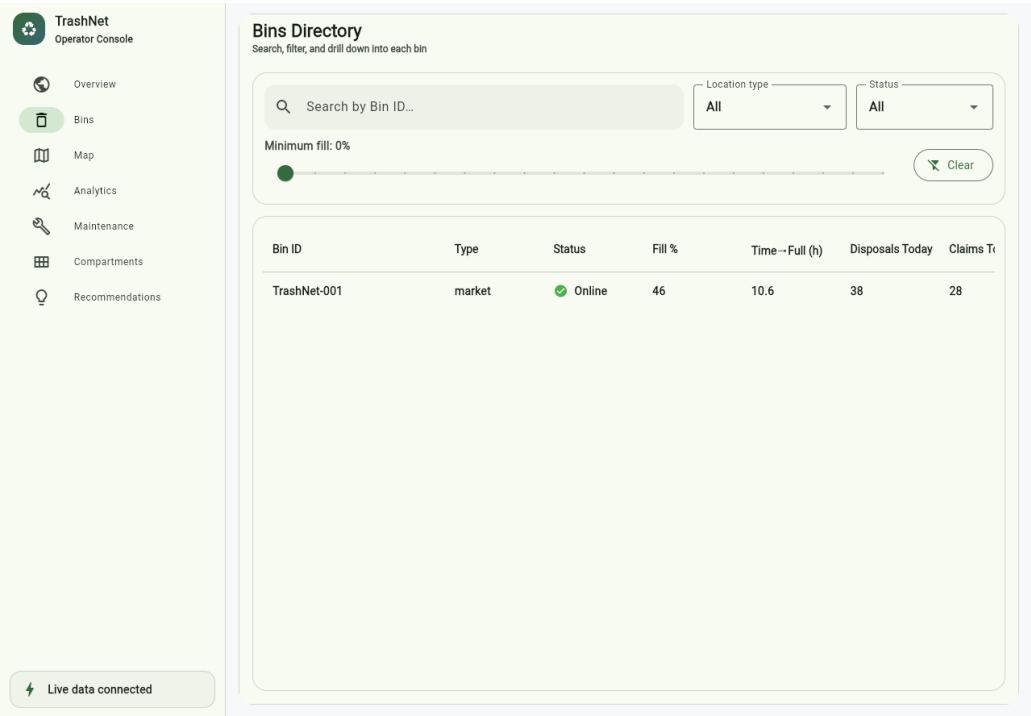
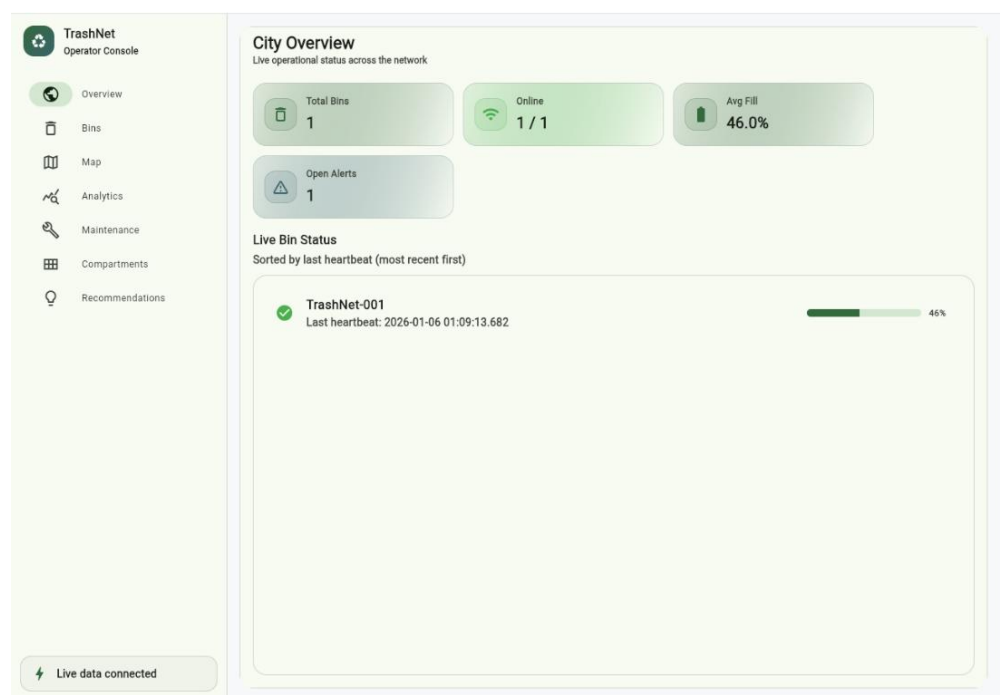
The system can be expanded for mass implementation across cities, public areas, and residential communities to improve overall waste management efficiency. Large-scale deployment would help reduce landfill waste, promote proper segregation at the source, and support cleaner urban environments. It would also enable centralized monitoring of multiple bins, making waste collection more organized and efficient.

REFERENCES

- [1] https://www.researchgate.net/figure/Organic-waste-classification-and-examples-Only-the-main-organic-waste-used-in_fig1_373868401
- [2] <https://devpost.com/software/leav>
- [3] R. Khadka, P. Joshi, N. K. Kanth, and S. Gwachha, "Waste Segregation System", *Int. J. Engin. Technol.*, vol. 2, no. 2, pp. 36–43, May 2025.
- [4] R. Sindhu, S. Vidhya, K. Rajat, S. Rohit, and S. Rachana, "Waste segregation using artificial intelligence," *International Journal of Scientific & Technology Research*, vol. 8, no. 12, pp. 903-905, 2019.
- [5] A. Okubanjo, O. Bashir Olufemi, A. Okandeji, and E. Daniel, "Smart Bin and IoT: A Sustainable Future for Waste Management System in Nigeria", *Gazi University Journal of Science*, vol. 37, no. 1, pp. 222–235, 2024, doi: 10.35378/gujs.1254271.
- [6] A. Singh and P. Tiwari, "Smart dustbin using iot notification",, 2021. <https://doi.org/10.21203/rs.3.rs-699249/v1>
- [7] K. Lam, N. Huynh, N. Ngoc, T. T. H. Nhu, N. Thao, P. Hao, V. Kiet, B. Huynh, and J. Kalita, "Using artificial intelligence and IoT for constructing a smart trash bin," in *Emerging Technologies in Data Mining and Information Security: Proceedings of IEMIS 2020, Volume 2*, Singapore: Springer, 2021, pp. 345–355, doi: 10.1007/978-981-16-8062-5_29.
- [8] A. Alourani, M. U. Ashraf, and M. Aloraini, "Smart waste management and classification system using advanced IoT and AI technologies," *PeerJ Computer Science*, vol. 11, e2777, 2025, doi: 10.7717/peerj-cs.2777.
- [9] M. S. Nafiz, S. S. Das, M. K. Morol, A. Al Juabir and D. Nandi, "ConvoWaste: An Automatic Waste Segregation Machine Using Deep Learning," *2023 3rd International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST)*, Dhaka, Bangladesh, 2023, pp. 181-186, doi: 10.1109/ICREST57604.2023.10070078.

- [10] Sundaralingam, S. and Ramanathan, N. 2023. A Deep Learning-Based approach to Segregate Solid Waste Generated in Residential Areas. *Engineering, Technology & Applied Science Research*. 13, 2 (Apr. 2023), 10439–10446. DOI: <https://doi.org/10.48084/etasr.5716>.
- [11] S. Poudel and P. Poudyal, “Classification of waste materials using CNN based on transfer learning,” in Proceedings of the 14th Annual Meeting of the Forum for Information Retrieval Evaluation (FIRE '22), New York, NY, USA, 2023, pp. 29–33. doi: 10.1145/3574318.3574345.
- [12] Algorimage Avijeet Biswal,"CNN in Deep Learning: Algorithm and Machine Learning"<https://www.simplilearn.com/tutorials/deeplearningtutorial/convolutional-neural-network>”
- [13] <https://www.geeksforgeeks.org/machine-learning/introduction-convolutional-neural-network/>

SCREENSHOTS



TrashNet

Operator Console

Overview

Bins

Map

Analytics

Maintenance

Compartments

Recommendations

⚡ Live data connected

← Bin – TrashNet-001

Live Status

History

Incidents

Collections

Display Name

New Road Gate Bin

Location Type

market

Status

Online

Fill Total

46%

Last Heartbeat

2026-01-06 01:09:13

Last Disposal

2026-01-06 01:09:13

Address

New Road Gate, Kathmandu

Fill by Compartment

PAPER

70%

BIO

10%

PLASTIC

35%

Open Alerts

overflow: 0

full: 0

damage: 1

offline: 0

TrashNet

Operator Console

Overview

Bins

Map

Analytics

Maintenance

Compartments

Recommendations

⚡ Live data connected

← Bin – TrashNet-001

Live Status

History

Incidents

Collections

qr_claim

2026-01-06 20:58:57

Fill: –%

qr_claim

2026-01-06 20:55:53

Fill: –%

qr_claim

2026-01-06 04:02:38

Fill: –%

qr_claim

2026-01-06 03:58:58

Fill: –%

qr_claim

2026-01-06 03:57:47

Fill: –%

qr_claim

2026-01-06 03:53:34

Fill: –%

disposal

2026-01-06 01:09:13

Fill: 46%

 TrashNet
Operator Console

Overview

Bins

Map

Analytics

Maintenance

Compartments

Recommendations

Live data connected

←

Bin – TrashNet-001

Live StatusHistoryIncidentsCollections

 lid_broken • medium • open
2026-01-06 01:09:13
Lid hinge damaged; still functional but needs repair.

 TrashNet
Operator Console

Overview

Bins

Map

Analytics

Maintenance

Compartments

Recommendations

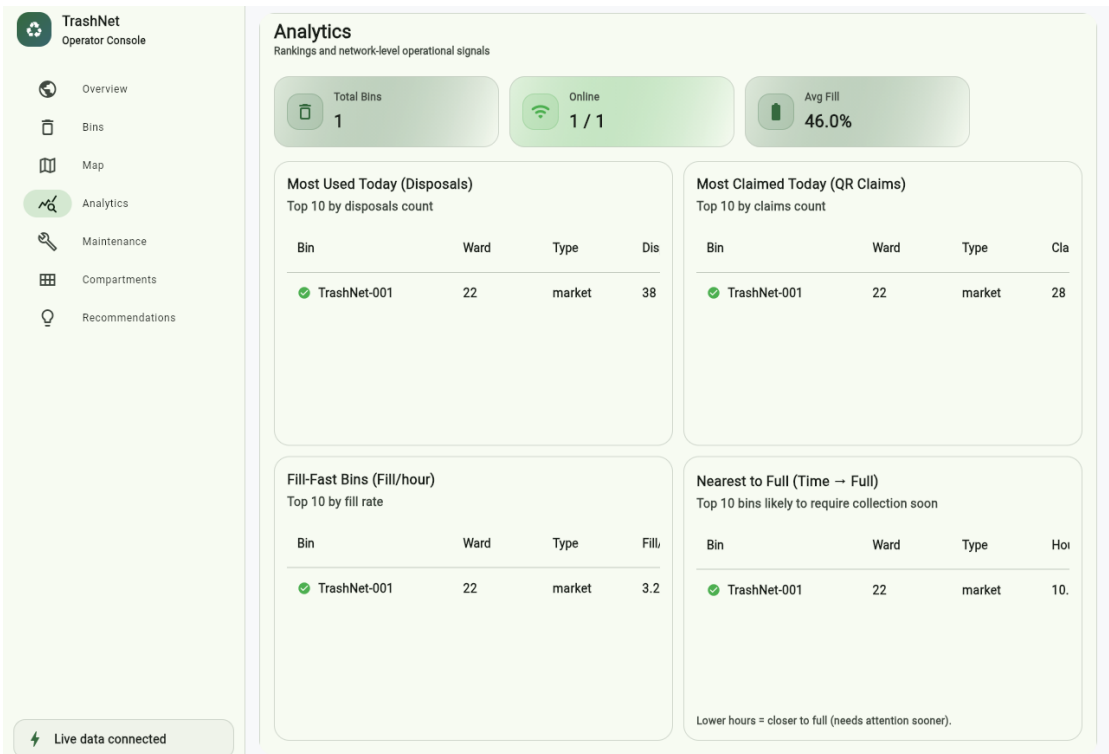
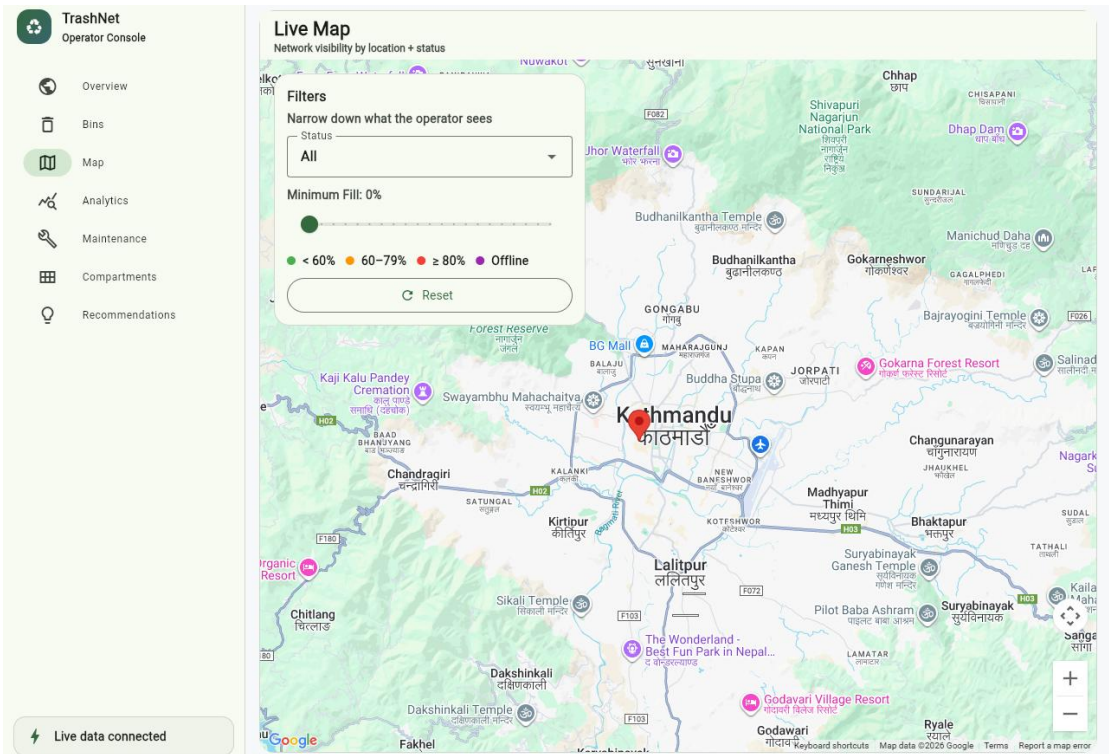
Live data connected

←

Bin – TrashNet-001

Live StatusHistoryIncidentsCollections

 Collected by Contractor-01
2026-01-06 01:09:13
Before: 92% → After: 8%
Morning collection completed.





TrashNet

Operator Console



Overview



Bins



Map



Analytics



Maintenance



Compartment



Recommendation



Live data connected

Maintenance

Damage & maintenance hotspots (last 30 days)

Hotspots

Bins with the highest maintenance burden

Bin ID	Incidents	Critical	High	Medium	Low	Avg Repair (hrs)
TrashNet-001	1	0	0	1	0	—



TrashNet

Operator Console



Overview



Bins



Map



Analytics



Maintenance



Compartment



Recommendation



Live data connected

Compartment

Which compartment will fill first? Use this to reconfigure capacity by area

Time To Full (TTF)

Sorted by earliest compartment to fill

Bin	Ward	Type	Online	Total%	Paper%	Paper/hr	Paper TTF(h)	Bio%	E
TrashNet-001	22	market		46	70	4.8	6.3	10	0

TTF is estimated using current % and fill-rate/hour. Use it to reconfigure compartment sizes by zone.



TrashNet

Operator Console

Overview

Bins

Map

Analytics

Maintenance

Compartment

Recommendations

Recommendations

Rule-based priority queue (higher score = needs attention sooner)

Priority Queue

Top 30 bins requiring attention

Priority	Bin	Ward	Type	Fill%	Disposals Today	QR Scan Ratio
High (52)	TrashNet-001	22	market	46	38	74%

Live data connected

